11-8-82

MIL-M-28787/122B(EC)
31 August 1982
SUPERSEDING
MIL-M-28787/122A(EC)
2 May 1979

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MULTIVIBRATOR, MONOSTABLE, DIGITAL, KEY CODE CDE

INACTIVE FOR NEW DESIGN AFTER DATE OF THIS SPECIFICATION

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic monostable multivibrator module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20360, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 1.9 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be CDE.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 75 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/122-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

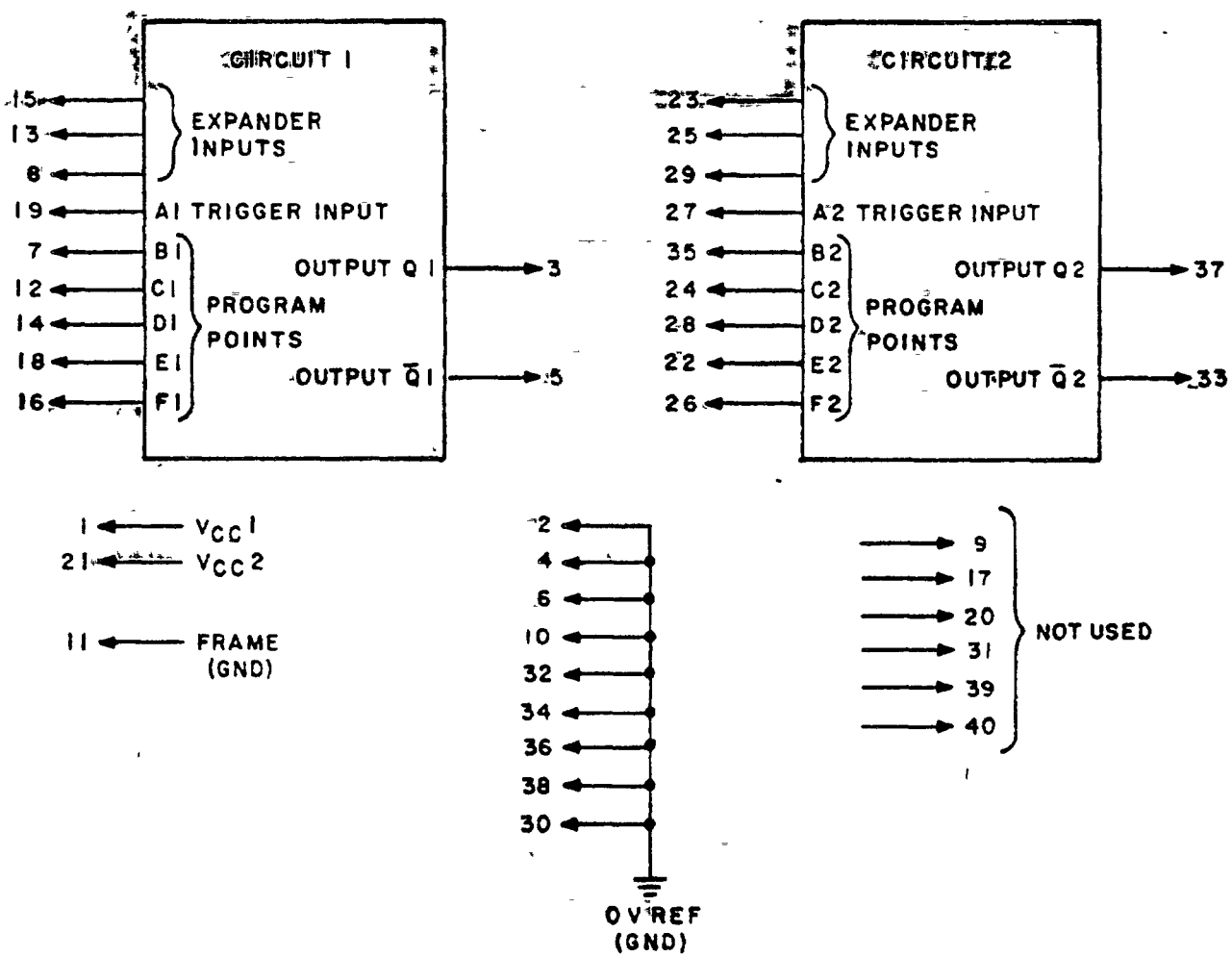
3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage ($V_{CC}$) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

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* FIGURE 1. CDE module function diagram, monostable multivibrator.



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* TABLE I. Module electrical requirements.

| Test condition | Parameter | Ref test table | Test type 1/ 2/ | Acceptance requirements 25°±5°C ambient temperature | | | | Operating temperature 0° to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| A1-A7 | Continuity | II | X | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | Ω |
| A8 | Continuity | II | S | 0 | 1 | 0 | 1 | 0 | 1 | 0 | 1 | Ω |
| B1,B2 | Isolation | II | X | 10 | ∞ | 10 | ∞ | 10 | ∞ | 10 | ∞ | MΩ |
| C1,C4, C5,C8 | Output voltage, low level | III | X | 0.0 | 0.4 | 0.00 | 0.45 | 0.0 | 0.4 | 0.00 | 0.45 | V |
| C2,C3,C6, C7 | Output voltage, high level | III | X | 3.8 | 5.5 | 3.5 | 5.5 | 3.8 | 5.5 | 3.5 | 5.5 | V |
| D1-D4 | Output short circuit current | IV | X | 16 | 65 | 16 | 65 | 16 | 65 | 16 | 65 | mA |
| D5, D7 | Input current, high level | IV | X | 0 | 4 | 0 | 8 | 0 | 6 | 0 | 10 | µA |
| D6, D8 | Input current, low level | IV | X | 0.0 | 3.2 | 0.0 | 3.2 | 0.0 | 3.2 | 0.0 | 3.2 | mA |
| E1 | Power supply current (quiescent) | V | S | 0 | 120 | 0 | 120 | 0 | 120 | 0 | 120 | mA |
| E2 | Power supply current (operating) | V | S | 0 | 120 | 0 | 120 | 0 | 120 | 0 | 120 | mA |
| F1-F4 | Output pulse width (1,000 µs) | VI | X | 930 | 1,070 | 910 | 1,090 | 900 | 1,100 | 880 | 1,120 | µs |
| F5-F8 | Output pulse width (200 µs) | VI | X | 186 | 214 | 182 | 218 | 180 | 220 | 176 | 224 | µs |
| F9-F12 | Output pulse width (300 µs) | VI | S | 279 | 321 | 273 | 327 | 270 | 330 | 264 | 336 | µs |
| F13-F16 | Output pulse width (400 µs) | VI | X | 372 | 428 | 364 | 436 | 360 | 440 | 352 | 448 | µs |

See footnotes at end of table.



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* TABLE I. Module electrical requirements - Continued.

| Test condition | Parameter | Ref test table | Test type 1/ 2/ | Acceptance requirements 25°±5°C ambient temperature | | | | Operating temperature 0° to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| F17-F20 | Output pulse width (500 µs) | VI | S | 465 | 535 | 455 | 545 | 450 | 550 | 440 | 560 | µs |
| F21-F24 | Output pulse width (600 µs) | VI | S | 558 | 642 | 546 | 654 | 540 | 660 | 528 | 672 | µs |
| F25-F28 | Output pulse width (700 µs) | VI | S | 651 | 749 | 637 | 763 | 630 | 770 | 616 | 784 | µs |
| F29-F32 | Output pulse width (800 µs) | VI | X | 744 | 856 | 728 | 872 | 720 | 880 | 704 | 896 | µs |
| F33-F36 | Output pulse width (900 µs) | VI | S | 837 | 963 | 819 | 981 | 810 | 990 | 792 | 1,008 | µs |
| F37-F40 | Output pulse width (1,000 µs) | VI | S | 930 | 1,070 | 910 | 1,090 | 900 | 1,100 | 880 | 1,120 | µs |
| F41-F44 | Output pulse width (1,100 µs) | VI | S | 1,023 | 1,177 | 1,001 | 1,200 | 990 | 1,210 | 968 | 1,232 | µs |
| F45-F48 | Output pulse width (1,200 µs) | VI | S | 1,116 | 1,284 | 1,092 | 1,308 | 1,080 | 1,320 | 1,056 | 1,344 | µs |
| F49-F52 | Output pulse width (1,300 µs) | VI | S | 1,209 | 1,390 | 1,183 | 1,417 | 1,170 | 1,430 | 1,144 | 1,456 | µs |
| F53-F56 | Output pulse width (1,400 µs) | VI | S | 1,302 | 1,498 | 1,274 | 1,526 | 1,260 | 1,540 | 1,232 | 1,568 | µs |
| F57-F60 | Output pulse width (1,500 µs) | VI | X | 1,395 | 1,605 | 1,365 | 1,635 | 1,350 | 1,650 | 1,320 | 1,680 | µs |
| G1-G5 | Transition time ($t_{THL}$) | VII | S | 3 | 40 | 3 | 45 | 3 | 40 | 3 | 45 | ns |
| G6-G10 | Transition time ($t_{TLH}$) | VII | S | 20 | 250 | 20 | 250 | 20 | 250 | 20 | 250 | ns |
| H1,H3,H5 | Propagation delay time ($t_{PLH}$) | VIII | S | 10 | 170 | 10 | 170 | 10 | 170 | 10 | 170 | ns |
| H2,H4,H6 | Propagation delay time ($t_{PHL}$) | VIII | S | 7 | 170 | 7 | 170 | 7 | 170 | 7 | 170 | ns |
| J1-J4 | Output pulse amplitude, low level | IX | S | 0 | 400 | 0 | 450 | 0 | 400 | 0 | 450 | mV |
| J5-J8 | Output pulse amplitude high level | IX | S | 3.8 | 5.5 | 3.5 | 5.5 | 3.8 | 5.5 | 3.5 | 5.5 | V |

See footnotes at end of table.



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* TABLE I. Module electrical requirements - Continued.

| Test condition | Parameter | Ref test table | Test type 1/ 2/ | Acceptance requirements 25°±5°C ambient temperature | | | | Operating temperature 0° to 60°C | | | | Unit |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Initial | | End-of-life | | Initial | | End-of-life | | |
| | | | | Min | Max | Min | Max | Min | Max | Min | Max | |
| K1-K4 | Output pulse, overshoot | X | S | 0 | 500 | 0 | 500 | 0 | 500 | 0 | 500 | mV |
| K5-K8 | Output pulse, undershoot | X | S | 0 | 500 | 0 | 500 | 0 | 500 | 0 | 500 | mV |
| K9-K12 | Output pulse, preshoot | X | S | 0 | 500 | 0 | 500 | 0 | 500 | 0 | 500 | mV |
| L1,L3 | Output pulse width (recovery time) | XI | S | 93 | 107 | 91 | 109 | 90 | 110 | 88 | 112 | µs |
| L2,L4 | Output pulse width (recovery time) | XI | S | 1,395 | 1,605 | 1,365 | 1,635 | 1,350 | 1,650 | 1,320 | 1,680 | µs |
| M1-M4 | Input trigger sensitivity | XII | S | 93 | 107 | 91 | 109 | 90 | 110 | 88 | 112 | µs |
| M5-M8 | Input immunity (20 ns) | XII | S | 0 | 5 | 0 | 5 | 0 | 5 | 0 | 5 | ns |
| M9-M12 | Input immunity (0.95 V) | XII | S | 0 | 5 | 0 | 5 | 0 | 5 | 0 | 5 | ns |
| N1,N2 | Power supply transient rise time | XIII | S | 9 | 53 | 9 | 53 | 9 | 53 | 9 | 53 | µs |
| N3,N4 | Power supply transient rise time | XIII | S | 1.5 | 40.0 | 1.5 | 40.0 | 1.5 | 40.0 | 1.5 | 40.0 | µs |
| N5-N8 | Power supply transient false trigger | XIII | S | 2.0 | 5.5 | 2.0 | 5.5 | 2.0 | 5.5 | 2.0 | 5.5 | V |
| P1-P12 | Crosstalk | XIV | S | 0 | 5 | 0 | 5 | 0 | 5 | 0 | 5 | ns |
| Q1,Q2 | Zener applied potential | XV | S | 3.8 | 5.5 | 3.5 | 5.5 | 3.8 | 5.5 | 3.5 | 5.5 | V |
| Q3,Q4 | Trigger voltage high | XV | S | 0 | 400 | 0 | 450 | 0 | 400 | 0 | 450 | mV |
| R1,R2 | $V_{OUT}$ (low loaded) | XVI | S | 0.0 | 1.1 | 0.0 | 1.1 | 0.0 | 1.1 | 0.0 | 1.1 | V |

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.



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4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

* 4.4.2 Order of testing. Test conditions A through R shall be performed in any order with the following exceptions. If test condition Q or R is part of a sequence of tests, the order of testing shall be as follows: Q, R where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

* 4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through R in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25° ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions A, C, D, E, F1-F16, F29-F32, F57-F60, G5, G10, H5, H6, M1-M4, and N shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25° ±5°C ambient.

* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests A and C through P shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests A through P shall be performed in the order specified in 4.4.2 using the 25° ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).



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* 6.3 Cross reference of classification. Military Specification MIL-M-28787/122 supersedes Department of the Navy, Naval Ordnance Systems Command Drawing 10001-2658711. Part number M28787/122-1 supersedes part numbers 2658711 and M28787/122.

* 6.4 Module description. The following electrical parameters reflect the 0° to 60°C end-of-life limits from table I:

| | |
|---|---|
| Supply voltage | 5.0 V dc ±10% |
| Maximum power dissipation | 660 mW (outputs not loaded) |
| Input voltage | |
| Low level | 0 V minimum, 0.95 V maximum |
| High level | 3.2 V minimum, $V_{CC}$ maximum |
| Input current | |
| Low level | 3.2 mA maximum |
| High level | 10.0 µA maximum |
| Output voltage | |
| High level | 3.5 V minimum, 5.5 V maximum |
| Low level | 0 V minimum, 0.45 V maximum |
| Q output drive current maximum capability | |
| Low level ($V_{OL}$ = 0.4 V) | 35.2 mA |
| High level ($V_{OH}$ = 3.8 V) | 200 µA |
| $\overline{Q}$ output drive current maximum capability | |
| Low level ($V_{OL}$ = 0.4 V) | 22.4 mA |
| High level ($V_{OH}$ = 3.8 V) | 200 µA |
| Output pulse width range | 100 µs to 1,500 µs in 100 µs increments |
| Input trigger pulse width | 85 ns minimum |
| Recovery time | |
| Without expander | 260 µs maximum |
| With expander | 2.8 ms maximum |

* 6.5 Module function. The module contains two monostable multivibrator (one-shot) circuits. For each input trigger pulse, each circuit delivers a positive output pulse Q and its complement pulse $\overline{Q}$ simultaneously. Each circuit is triggered on the trailing edge of a positive input pulse. Each circuit is capable of being driven by a source with an impedance of 12 kilohms or less when the input voltage level is high and an impedance of 100 ohms or less when the input voltage level is low.

6.5.1 Expander. The pulse width range of circuits 1 and 2 (100 to 1,500 µs) are expandable by a factor of 10 (1 to 15 ms) in steps of 1 ms. Circuit 1 is expanded when pins 8 and 15 are commoned, pins 13 and 16 are commoned, and a 0.503 µF capacitor is connected between pins 8 and 16. Circuit 2 is expanded when pins 23 and 26 are commoned, pins 25 and 29 are commoned, and a 0.503 µF capacitor is connected between pins 26 and 29. Circuits 1 and 2 are independently and simultaneously capable of having their pulse widths expanded by a factor of 10.

6.5.2 Output loading. The $\overline{Q}$ output is capable of driving 14 standard logic loads (22.4 mA) at a low level. The Q output is capable of driving 22 standard logic loads (35.2 mA) at a low level. Each output is capable of driving 40 standard logic loads (200 µA) at a high level.



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6.5.3 Parallel outputs. Paralleling of outputs is not permissible.

6.5.4 Input to output relationship. Each monostable multivibrator circuit has the input to output relationship shown on figure 1.

6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

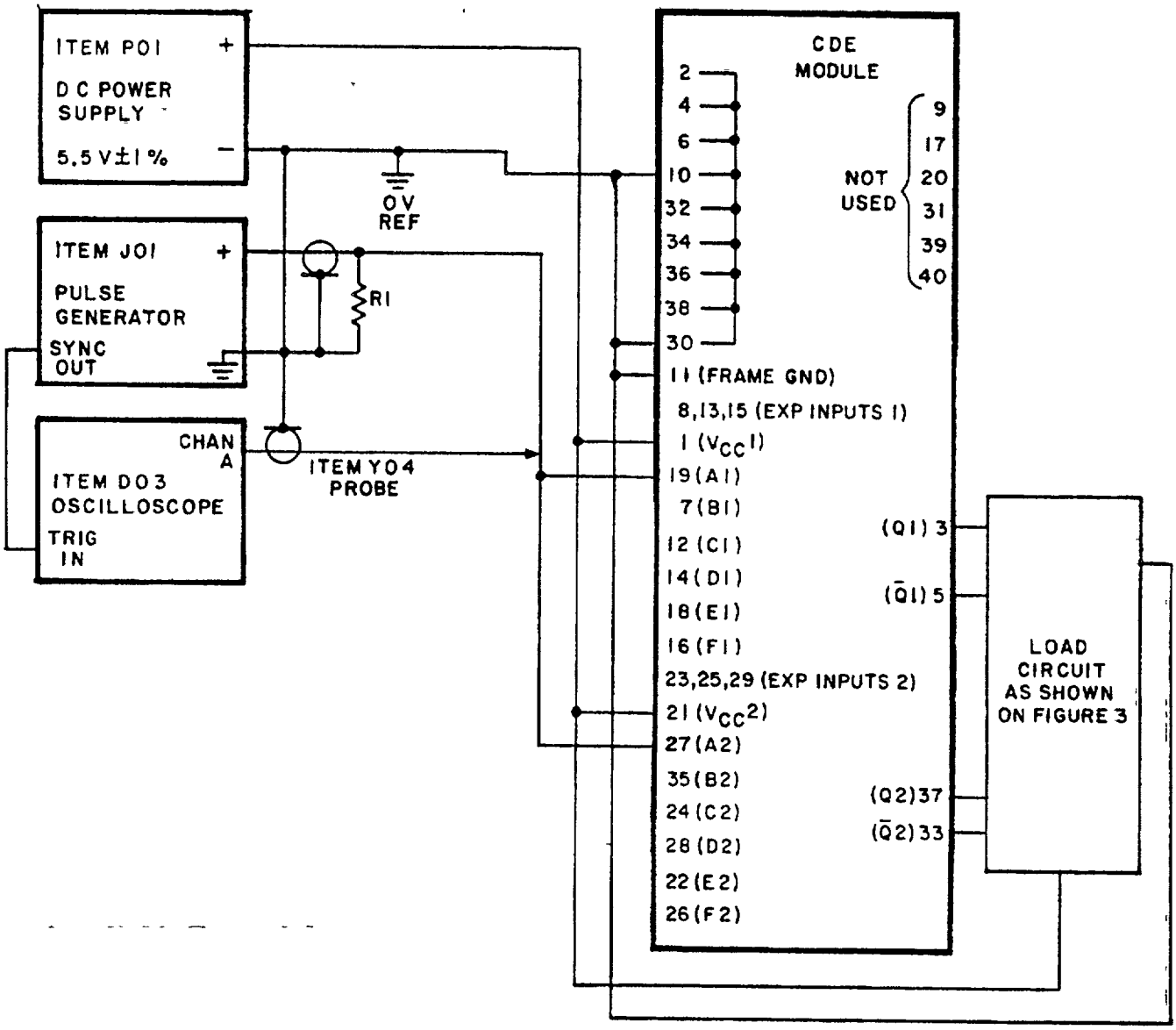
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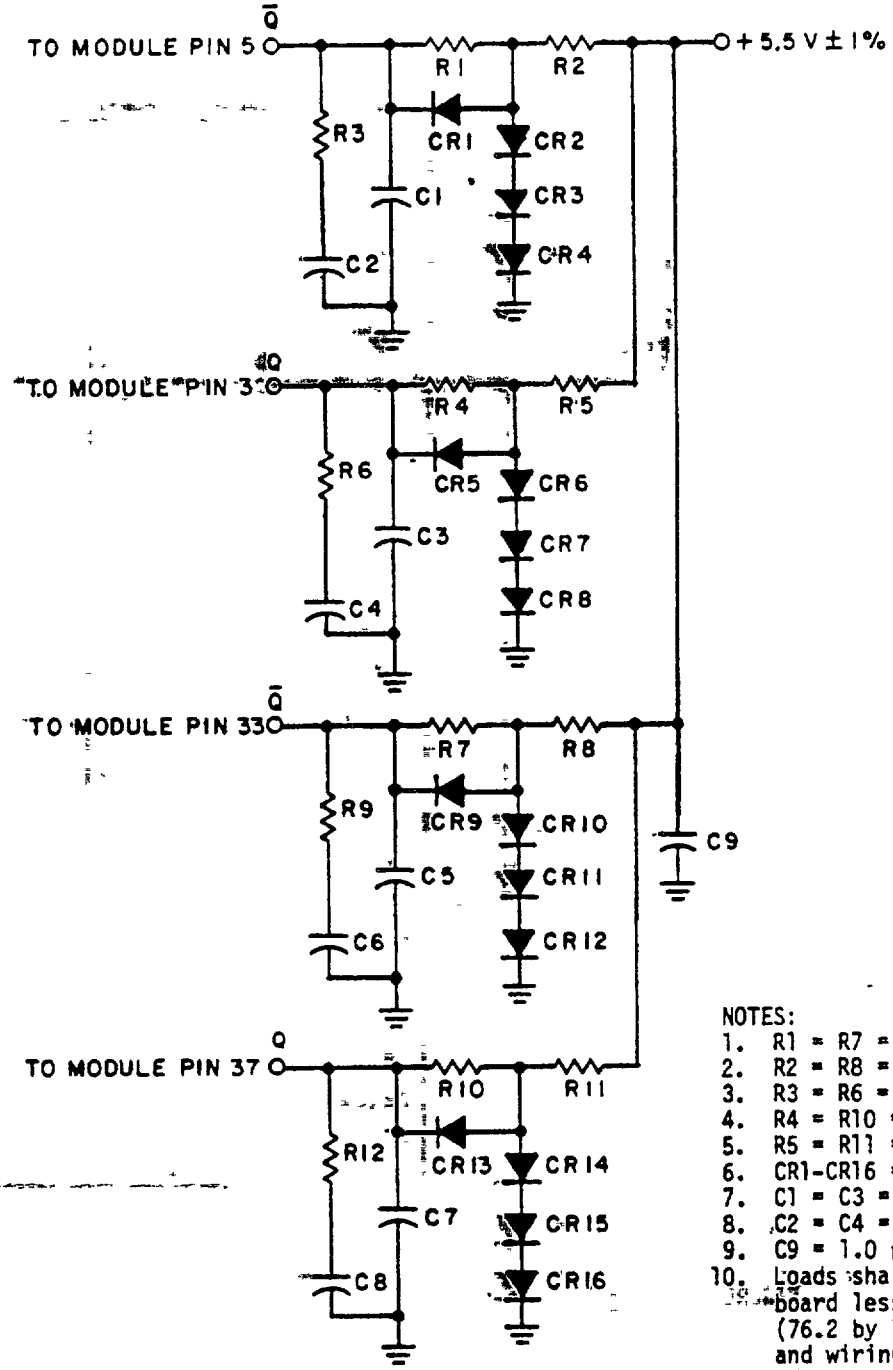



NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. Item J01 parameters:
   PA = 5 V ±5%.
   PW (1.5 V level) = 150 ns ±10%.
   $t_{TLH} = t_{THL}$ = 20 ns ±10%.
   PRP = 30 µs ±10%.

* FIGURE 2. Operating temperature, component temperature, and life test circuit.



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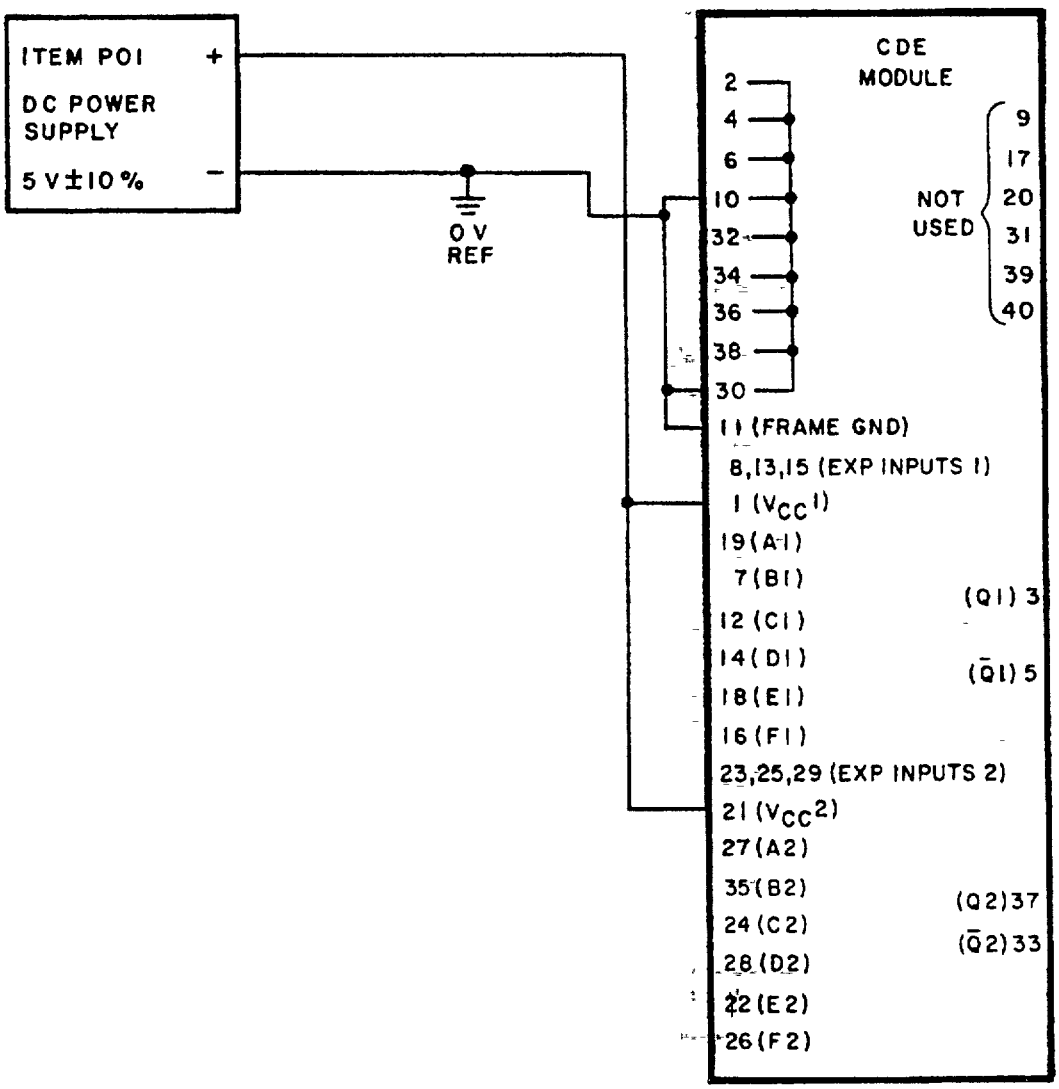



NOTES:
1. R1 = R7 = 7.5 kΩ ±1%, 1/8 W.
2. R2 = R8 = 215Ω ±1%, 1/8 W.
3. R3 = R6 = R9 = R12 = 100Ω ±1%, 1/8 W.
4. R4 = R10 = 7.15 kΩ ±1%, 1/8 W.
5. R5 = R11 = 140Ω ±1%, 1/8 W.
6. CR1-CR16 = 1N4150 or equivalent.
7. C1 = C3 = C5 = C7 = 33 pF ±10%.
8. C2 = C4 = C6 = C8 = 330 pF ±10%.
9. C9 = 1.0 µF ±10%.
10. Loads shall be assembled on a board less than 3 by 3 inches (76.2 by 76.2 mm) with components and wiring within 0.50 inch (12.7 mm) of the module under test.

* FIGURE 3. CDE module load circuit.



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* FIGURE 4. Vibration test circuit.



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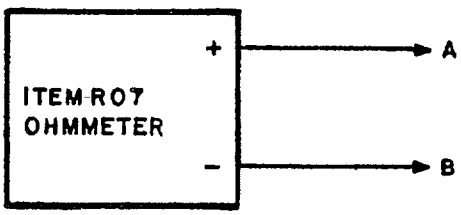
* TABLE II. Continuity and isolation test.

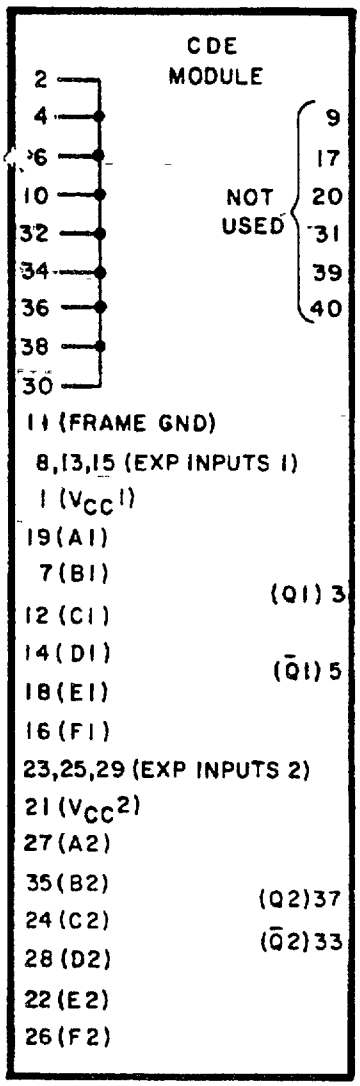
| Test condition | Parameter | Connect | | Initial limits 25°±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| A1 | Continuity | 10 | 2 | 0 | 1 | Ω | 5 |
| A2 | ↓ | 10 | 4 | ↓ | ↓ | ↓ | ↓ |
| A3 | | 10 | 6 | | | | |
| A4 | | 30 | 32 | | | | |
| A5 | | ↓ | 34 | | | | |
| A6 | | | 36 | | | | |
| A7 | | | 38 | | | | |
| A8 | ↓ | 11 | Frame | ↓ | ↓ | ↓ | ↓ |
| B1 | Isolation | 11 | All other pins interconnected | 10 | ∞ | MΩ | |
| B2 | Isolation | Unused pins 9,17,20,31, and 40 interconnected | To all other pins interconnected | 10 | ∞ | MΩ | |





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NOTE: For test condition A8, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

* FIGURE 5. Continuity and isolation test circuit for table II.



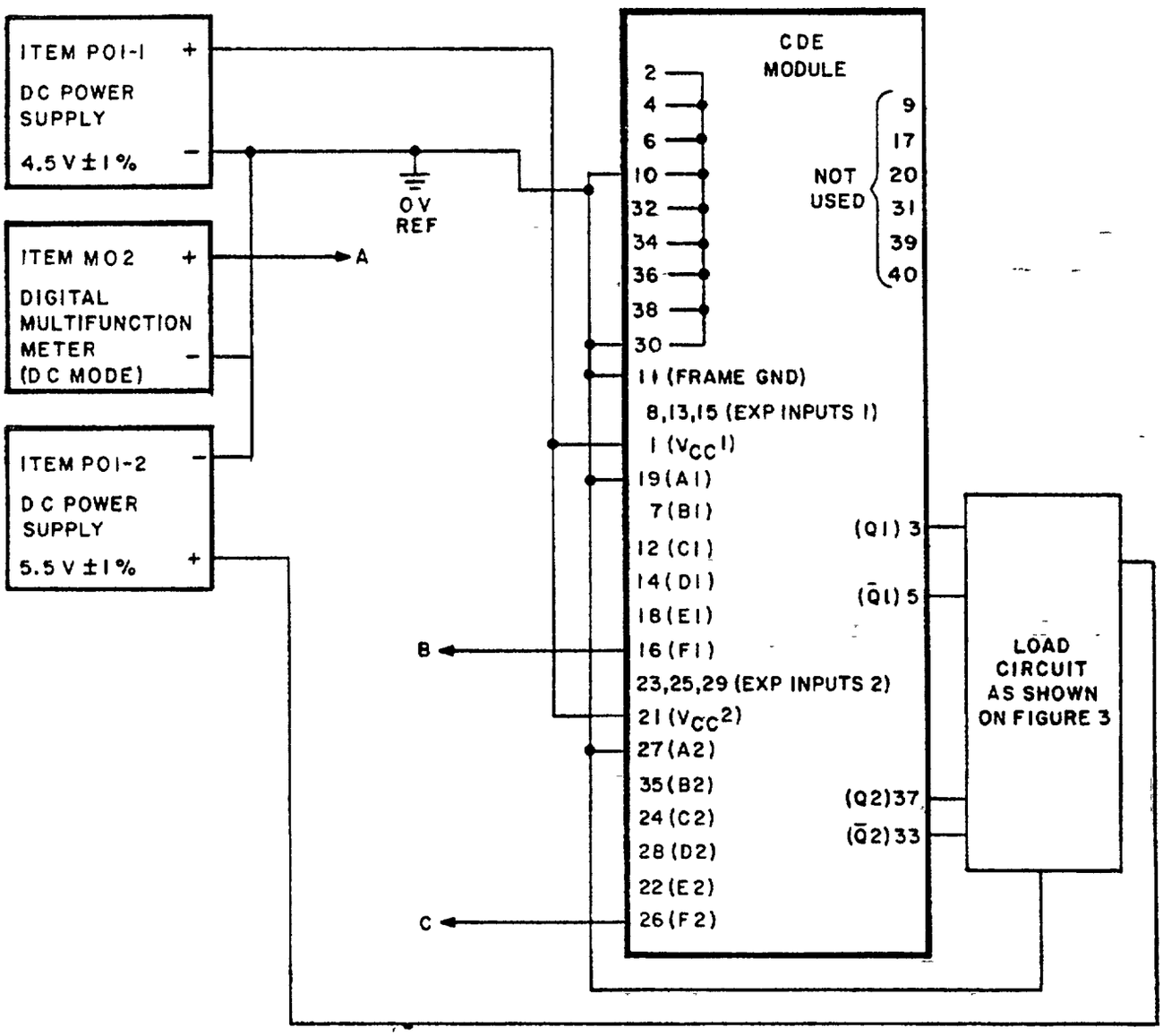
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* TABLE III. Output voltage (low-level and high-level) test.

| Test condition | Parameter | Connect | | | Initial limits 25°±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | A-to-pin | B-to-pin | C-to-pin | Min | Max | | |
| C1 | Q1 Output voltage, low level | 3 | Open | Open | 0.0 | 0.4 | V | 6 |
| C2 | $\overline{Q1}$ Output voltage, high level | 5 | Open | ↓ | 3.8 | 5.5 | ↓ | ↓ |
| C3 | Q1 Output voltage, high level | 3 | 10 | ↓ | 3.8 | 5.5 | ↓ | ↓ |
| C4 | $\overline{Q1}$ Output voltage, low level | 5 | 10 | ↓ | 0.0 | 0.4 | ↓ | ↓ |
| C5 | Q2 Output voltage, low level | 37 | Open | ↓ | 0.0 | 0.4 | ↓ | ↓ |
| C6 | $\overline{Q2}$ Output voltage, high level | 33 | ↓ | ↓ | 3.8 | 5.5 | ↓ | ↓ |
| C7 | Q2 Output voltage, high level | 37 | ↓ | 30 | 3.8 | 5.5 | ↓ | ↓ |
| C8 | $\overline{Q2}$ Output voltage, low level | 33 | ↓ | 30 | 0.0 | 0.4 | ↓ | ↓ |



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NOTES:
1. Prior to making measurements for test conditions C3, C4, C7 and C8, it may be necessary to provide additional stimuli to the module to force it into the desired quiescent state.
2. Interconnecting conductor length between pins 10 and 16 and between pins 26 and 30 shall not exceed 6 inches (152.4 mm).

* FIGURE 6. Output voltage (low-level and high-level) test circuit for table III.

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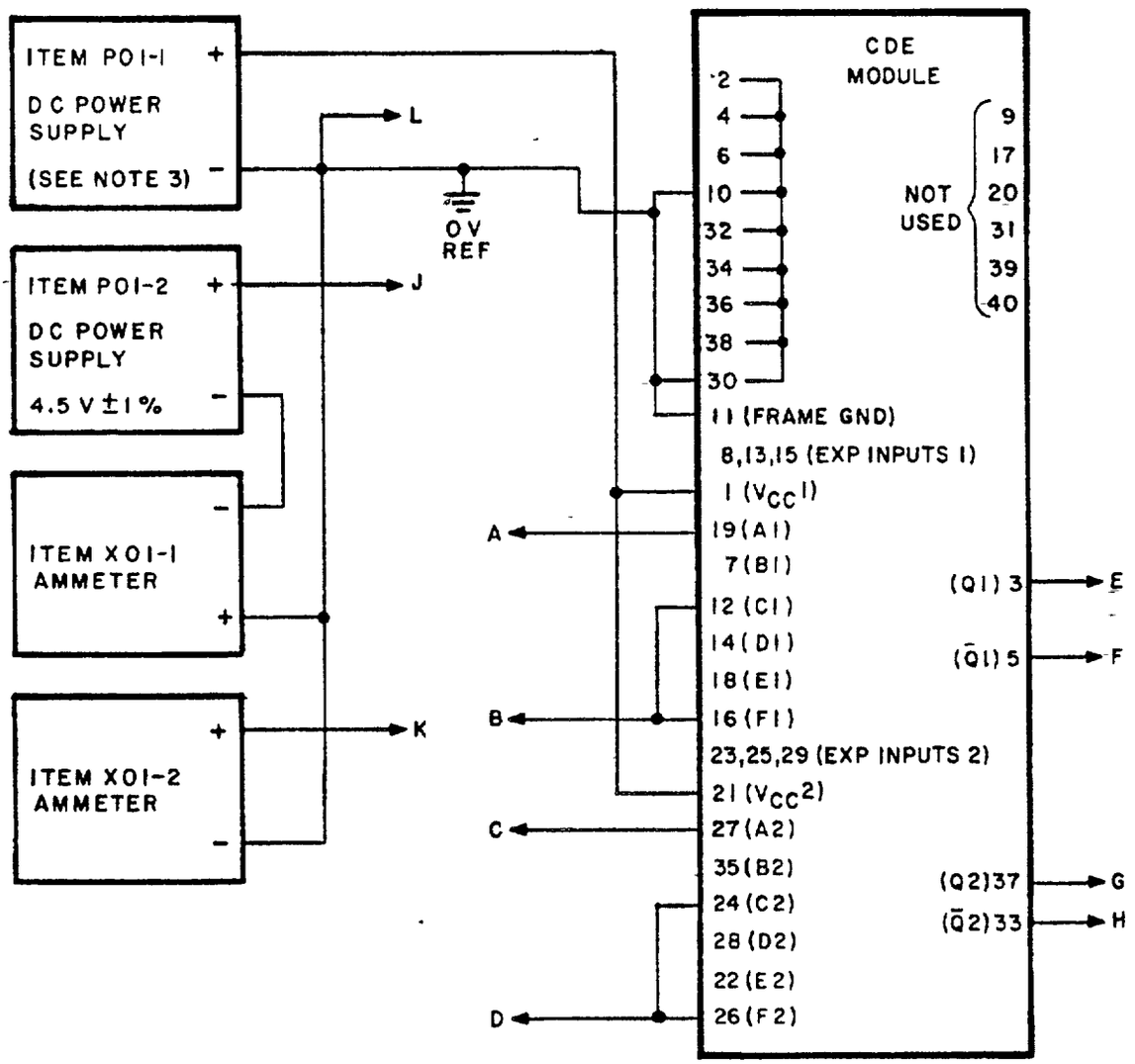
* TABLE IV. Output short circuit current and input current (high-level and low-level) test.

| Test condition | Parameter | Connect | | | | | | | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | E to pin | F to pin | G to pin | H to pin | Min | Max | | |
| D1 | Q1 Output short circuit current | L | 10, 12 | L | 24 | K | Open | Open | Open | 16 | 65 | mA | 7 |
| D2 | $\overline{Q1}$ | | 12 | | 24 | Open | K | Open | Open | | | | |
| D3 | Q2 | | | | 24, 30 | | Open | K | Open | | | | |
| D4 | $\overline{Q2}$ | | | | 24 | | | Open | K | | | | |
| D5 | C1 and C2 Input current, high level | J | | Open | | | | | Open | 0 | 4 | µA | |
| D6 | C1 and C2 Input current, low level | K | | Open | | | | | | 0.0 | 3.2 | mA | |
| D7 | C1 and C2 Input current, high level | Open | | J | | | | | | 0 | 4 | µA | |
| D8 | C1 and C2 Input current, low level | Open | | K | | | | | | 0.0 | 3.2 | mA | |





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CAUTION:

The chassis of power supply PO1-2 shall be isolated from earth ground.

CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTES:
1. Prior to making measurements for test conditions D1 and D3, it may be necessary to provide additional stimuli to the module to force it into the desired quiescent state.
2. The conductor lengths between any two pins listed in the CONNECT column of table IV shall not exceed 6 inches (152.4 mm).
3. $V_{CC}$ = 5.5 V ±1% for all test conditions except for test conditions D5 and D7 where $V_{CC}$ = 4.5 V ±1%.

* FIGURE 7. Output short circuit current and input current (high-level and low-level) test circuit for table IV.



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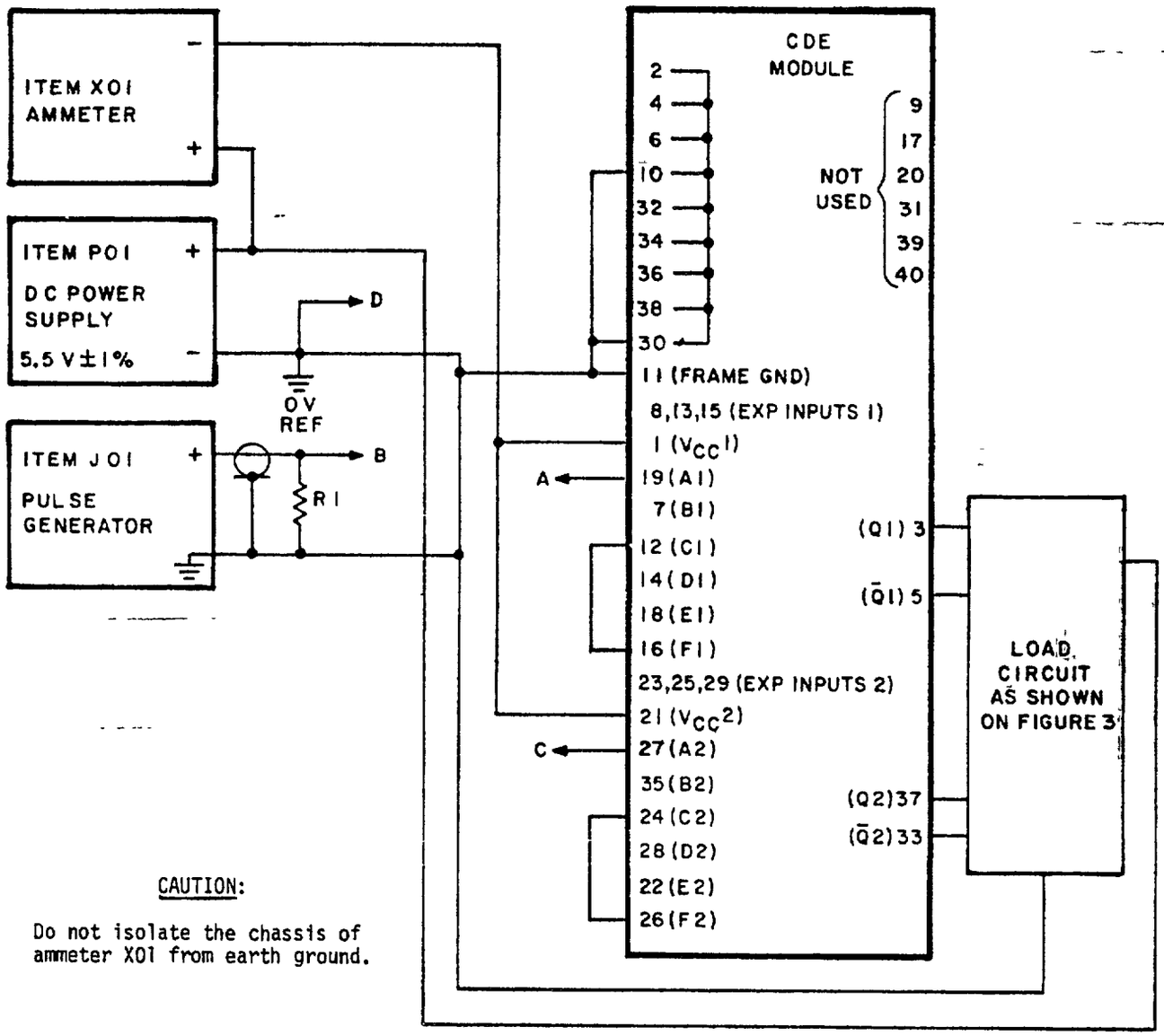
* TABLE V. Power supply current test.

| Test condition | Parameter | Connect | | Initial limits 25°±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | C to pin | Min | Max | | |
| E1 | Power supply current (quiescent) | D | D | 0 | 120 | mA | 8 |
| E2 | Power supply current (operating) | B | B | 0 | 120 | mA | 8 |





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CAUTION:

Do not isolate the chassis of ammeter XO1 from earth ground.

NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. Item JO1 parameters:
   PA = 4 V ±5% with reference to 0 V.
   PW (1.5 V level) = 125 ns ±10%.
   $t_{TLH} = t_{THL}$ = 20 ns ±10%.
   PRP = 400 µs ±10%.

* FIGURE 8. Power supply current test circuit for table V.



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* TABLE VI. Output pulse width test.

| Test condition | Parameter | $V_{CC}$ | Connect | | | | | | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | D to pin | G to pin | H to pin | J to pin | Min. | Max | | |
| F1 | Q1 Output pulse width (1,000 µs) | 5.5 | 12 | 16 | Open | Open | 8,15, 19 | Open | 3,13, 16 | 930 | 1,070 | µs | 9,10 |
| F2 | $\overline{Q1}$ Output pulse width (1,000 µs) | 4.5 | 12 | 16 | ↓ | ↓ | 8,15, 19 | ↓ | 5,13, 16 | ↓ | ↓ | ↓ | ↓ |
| F3 | Q2 Output pulse width (1,000 µs) | 4.5 | 24 | 26 | ↓ | ↓ | 23,26, 27 | ↓ | 25,29, 37 | ↓ | ↓ | ↓ | ↓ |
| F4 | $\overline{Q2}$ Output pulse width (1,000 µs) | 5.5 | 24 | 26 | ↓ | ↓ | 23,26, 27 | ↓ | 25,29, 33 | ↓ | ↓ | ↓ | ↓ |
| F5 | Q1 Output pulse width (200 µs) | 5.5 | 7 | 12 | 14 | 16 | 19 | 3 | Open | 186 | 214 | ↓ | ↓ |
| F6 | $\overline{Q1}$ Output pulse width (200 µs) | 4.5 | 7 | 12 | 14 | 16 | 19 | 5 | ↓ | ↓ | ↓ | ↓ | ↓ |
| F7 | Q2 Output pulse width (200 µs) | 4.5 | 24 | 35 | 26 | 28 | 27 | 37 | ↓ | ↓ | ↓ | ↓ | ↓ |
| F8 | $\overline{Q2}$ Output pulse width (200 µs) | 5.5 | 24 | 35 | 26 | 28 | 27 | 33 | ↓ | ↓ | ↓ | ↓ | ↓ |
| F9 | Q1 Output pulse width (300 µs) | 4.5 | 14 | 16 | Open | Open | 19 | 3 | ↓ | 279 | 321 | ↓ | ↓ |
| F10 | $\overline{Q1}$ Output pulse width (300 µs) | 5.5 | 14 | 16 | ↓ | ↓ | 19 | 5 | ↓ | ↓ | ↓ | ↓ | ↓ |
| F11 | Q2 Output pulse width (300 µs) | 5.5 | 26 | 28 | ↓ | ↓ | 27 | 37 | ↓ | ↓ | ↓ | ↓ | ↓ |
| F12 | $\overline{Q2}$ Output pulse width (300 µs) | 4.5 | 26 | 28 | ↓ | ↓ | 27 | 33 | ↓ | ↓ | ↓ | ↓ | ↓ |



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* TABLE VI. Output pulse width test - Continued.

| Test condition | Parameter | $V_{CC}$ | Connect | | | | | | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | D to pin | G to pin | H to pin | J to pin | Min | Max | | |
| F13 | Q1 Output pulse width (400 µs) | 5.5 | 7 | 14 | 16 | 18 | 19 | 3 | Open | 372 | 428 | µs | 9,10 |
| F14 | $\overline{Q1}$ Output pulse width (400 µs) | 4.5 | 7 | 14 | 16 | 18 | 19 | 5 | | | | | |
| F15 | Q2 Output pulse width (400 µs) | 4.5 | 28 | 35 | 22 | 26 | 27 | 37 | | | | | |
| F16 | $\overline{Q2}$ Output pulse width (400 µs) | 5.5 | 28 | 35 | 22 | 26 | 27 | 33 | | | | | |
| F17 | Q1 Output pulse width (500 µs) | 4.5 | 12 | 14 | 16 | 18 | 19 | 3 | | 465 | 535 | | |
| F18 | $\overline{Q1}$ Output pulse width (500 µs) | 5.5 | 12 | 14 | 16 | 18 | 19 | 5 | | | | | |
| F19 | Q2 Output pulse width (500 µs) | 5.5 | 24 | 28 | 22 | 26 | 27 | 37 | | | | | |
| F20 | $\overline{Q2}$ Output pulse width (500 µs) | 4.5 | 24 | 28 | 22 | 26 | 27 | 33 | | | | | |
| F21 | Q1 Output pulse width (600 µs) | 4.5 | 7 | 12 | 16 | 18 | 19 | 3 | | 558 | 642 | | |
| F22 | $\overline{Q1}$ Output pulse width (600 µs) | 5.5 | 7 | 12 | 16 | 18 | 19 | 5 | | | | | |
| F23 | Q2 Output pulse width (600 µs) | 5.5 | 24 | 35 | 22 | 26 | 27 | 37 | | | | | |
| F24 | $\overline{Q2}$ Output pulse width (600 µs) | 4.5 | 24 | 35 | 22 | 26 | 27 | 33 | | | | | |





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* TABLE VI. Output pulse width test - Continued.

| Test condition | Parameter | $V_{CC}$ | Connect | | | | | | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | D to pin | G to pin | H to pin | J to pin | Min | Max | | |
| F25 | Q1 Output pulse width (700 µs) | 4.5 | 16 | 18 | Open | Open | 19 | 3 | Open | 651 | 749 | µs | 9,10 |
| F26 | $\overline{Q1}$ Output pulse width (700 µs) | 5.5 | 16 | 18 | | | 19 | 5 | | | | | |
| F27 | Q2 Output pulse width (700 µs) | 5.5 | 22 | 26 | | | 27 | 37 | | | | | |
| F28 | $\overline{Q2}$ Output pulse width (700 µs) | 4.5 | 22 | 26 | | | 27 | 33 | | | | | |
| F29 | Q1 Output pulse width (800 µs) | 5.5 | 7 | 18 | | | 19 | 3 | | 744 | 856 | | |
| F30 | $\overline{Q1}$ Output pulse width | 4.5 | 7 | 18 | | | 19 | 5 | | | | | |
| F31 | Q2 Output pulse width (800 µs) | 4.5 | 22 | 35 | | | 27 | 37 | | | | | |
| F32 | $\overline{Q2}$ Output pulse width (800 µs) | 5.5 | 22 | 35 | | | 27 | 33 | | | | | |
| F33 | Q1 Output pulse width (900 µs) | 4.5 | 12 | 18 | | | 19 | 3 | | 837 | 963 | | |
| F34 | $\overline{Q1}$ Output pulse width (900 µs) | 5.5 | 12 | 18 | | | 19 | 5 | | | | | |
| F35 | Q2 Output pulse width (900 µs) | 5.5 | 22 | 24 | | | 27 | 37 | | | | | |
| F36 | $\overline{Q2}$ Output pulse width (900 µs) | 4.5 | 22 | 24 | | | 27 | 33 | | | | | |



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* TABLE VI. Output pulse width test - Continued.

| Test condition | Parameter | $V_{CC}$ | Connect | | | | | | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | D to pin | G to pin | H to pin | J to pin | Min | Max | | |
| F37 | Q1 Output pulse width (1,000 µs) | 4.5 | 14 | 18 | 7 | 12 | 19 | 3 | Open | 930 | 1,070 | µs | 9,10 |
| F38 | $\overline{Q1}$ Output pulse width (1,000 µs) | 5.5 | 14 | 18 | 7 | 12 | 19 | 5 | | | | | |
| F39 | Q2 Output pulse width (1,000 µs) | 5.5 | 24 | 35 | 22 | 28 | 27 | 37 | | | | | |
| F40 | $\overline{Q2}$ Output pulse width (1,000 µs) | 4.5 | 24 | 35 | 22 | 28 | 27 | 33 | | | | | |
| F41 | Q1 Output pulse width (1,100 µs) | 4.5 | 14 | 18 | Open | Open | 19 | 3 | | 1,023 | 1,177 | | |
| F42 | $\overline{Q1}$ Output pulse width (1,100 µs) | 5.5 | 14 | 18 | | | 19 | 5 | | | | | |
| F43 | Q2 Output pulse width (1,100 µs) | 5.5 | 22 | 28 | | | 27 | 37 | | | | | |
| F44 | $\overline{Q2}$ Output pulse width (1,100 µs) | 4.5 | 22 | 28 | | | 27 | 33 | | | | | |
| F45 | Q1 Output pulse width (1,200 µs) | 4.5 | 7 | 14 | | | 19 | 3 | | 1,116 | 1,284 | | |
| F46 | $\overline{Q1}$ Output pulse width (1,200 µs) | 5.5 | 7 | 14 | | | 19 | 5 | | | | | |
| F47 | Q2 Output pulse width (1,200 µs) | 5.5 | 28 | 35 | | | 27 | 37 | | | | | |
| F48 | $\overline{Q2}$ Output pulse width (1,200 µs) | 4.5 | 28 | 35 | | | 27 | 33 | | | | | |





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* TABLE VI. Output pulse width test - Continued.

| Test condition | Parameter | $V_{CC}$ | Connect | | | | | | | Initial limits 25° ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | A to pin | B to pin | C to pin | D to pin | G to pin | H to pin | J to pin | Min | Max | | |
| F49 | Q1 Output pulse width (1,300 µs) | 4.5 | 12 | 14 | Open | Open | 19 | 3 | Open | 1,209 | 1,390 | µs | 9,10 |
| F50 | $\overline{Q1}$ Output pulse width (1,300 µs) | 5.5 | 12 | 14 | | | 19 | 5 | | | | | |
| F51 | Q2 Output pulse width (1,300 µs) | 5.5 | 24 | 28 | | | 27 | 37 | | | | | |
| F52 | $\overline{Q2}$ Output pulse width (1,300 µs) | 4.5 | 24 | 28 | | | 27 | 33 | | | | | |
| F53 | Q1 Output pulse width (1,400 µs) | 4.5 | 7 | 12 | | | 19 | 3 | | 1,302 | 1,498 | | |
| F54 | $\overline{Q1}$ Output pulse width (1,400 µs) | 5.5 | 7 | 12 | | | 19 | 5 | | | | | |
| F55 | Q2 Output pulse width (1,400 µs) | 5.5 | 24 | 35 | | | 27 | 37 | | | | | |
| F56 | $\overline{Q2}$ Output pulse width (1,400 µs) | 4.5 | 24 | 35 | | | 27 | 33 | | | | | |
| F57 | Q1 Output pulse width (1,500 µs) | 5.5 | Open | Open | | | 19 | 3 | | 1,395 | 1,605 | | |
| F58 | $\overline{Q1}$ Output pulse width (1,500 µs) | 4.5 | | | | | 19 | 5 | | | | | |
| F59 | Q2 Output pulse width (1,500 µs) | 4.5 | | | | | 27 | 37 | | | | | |
| F60 | $\overline{Q2}$ Output pulse width (1,500 µs) | 5.5 | | | | | 27 | 33 | | | | | |



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* TABLE VII. Transition time ($t_{THL}$ and $t_{TLH}$) test.

| Test condition | Parameter | Connect 1/ E to pin | F to pin | G to pin | H to pin | J to pin | Initial limits 25°±5°C ambient Min | Max | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| G1 | Q1 Transition time ($t_{THL}$) | Open | Open | 19 | 3 | Open | 3 | 40 | ns | 9,10 |
| G2 | $\overline{Q1}$ | | | 19 | 5 | | | | | |
| G3 | Q2 | | | 27 | 37 | | | | | |
| G4 | $\overline{Q2}$ | | | 27 | 33 | | | | | |
| G5 | $\overline{Q2}$ | 23,26 | 25,29 | 27 | Open | 33 | | | | |
| G6 | Q1 Transition time ($t_{TLH}$) | Open | Open | 19 | 3 | Open | 20 | 250 | | |
| G7 | $\overline{Q1}$ | | | 19 | 5 | | | | | |
| G8 | Q2 | | | 27 | 37 | | | | | |
| G9 | $\overline{Q2}$ | | | 27 | 33 | | | | | |
| G10 | Q1 | 8,15 | 13,16 | 19 | Open | 3 | | | | |

1/ Connect pin 12 to pin 16 and pin 24 to pin 26 for all test conditions, except test conditions G5 and G10.





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* TABLE VIII. Propagation delay time ($t_{PLH}$ and $t_{PHL}$) test.

| Test condition | Parameter | Connect 1/ | | | | | Initial limits 25°±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|---|
| | | E to pin | F to pin | G to pin | H to pin | J to pin | Min | Max | | |
| H1 | Q1 Propagation delay time ($t_{PLH}$) | Open | Open | 19 | 3 | Open | 10 | 170 | ns | 9;10 |
| H2 | $\overline{Q1}$ Propagation delay time ($t_{PHL}$) | | | 19 | 5 | | 7 | | | |
| H3 | Q2 Propagation delay time ($t_{PLH}$) | | | 27 | 37 | | 10 | | | |
| H4 | $\overline{Q2}$ Propagation delay time ($t_{PHL}$) | | | 27 | 33 | | 7 | | | |
| H5 | Q1 Propagation delay time ($t_{PLH}$) | 8,15 | 13,16 | 19 | Open | 3 | 10 | | | |
| H6 | $\overline{Q2}$ Propagation delay time ($t_{PHL}$) | 23,26 | 25,29 | 27 | Open | 33 | 7 | | | |

1/ Connect pin 12 to pin 16 and pin 24 to pin 26 for all test conditions, except test conditions H5 and $\overline{H6}$.



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* TABLE IX. Output pulse amplitude (low-level and high-level) test.

| Test condition | Parameter | Connect 1/ | | Initial limits 25°±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | G to pin | J to pin | Min | Max | | |
| J1 | Q1 Output pulse amplitude, low level | 19 | 3 | 0 | 400 | mV | 9,10 |
| J2 | $\overline{Q1}$ | 19 | 5 | | | | |
| J3 | Q2 | 27 | 37 | | | | |
| J4 | $\overline{Q2}$ | 27 | 33 | | | | |
| J5 | Q1 Output pulse amplitude, high level | 19 | 3 | 3.8 | 5.5 | V | |
| J6 | $\overline{Q1}$ | 19 | 5 | | | | |
| J7 | Q2 | 27 | 37 | | | | |
| J8 | $\overline{Q2}$ | 27 | 33 | | | | |

1/ Connect pin 12 to pin 16 and pin 24 to pin 26 for all test conditions.





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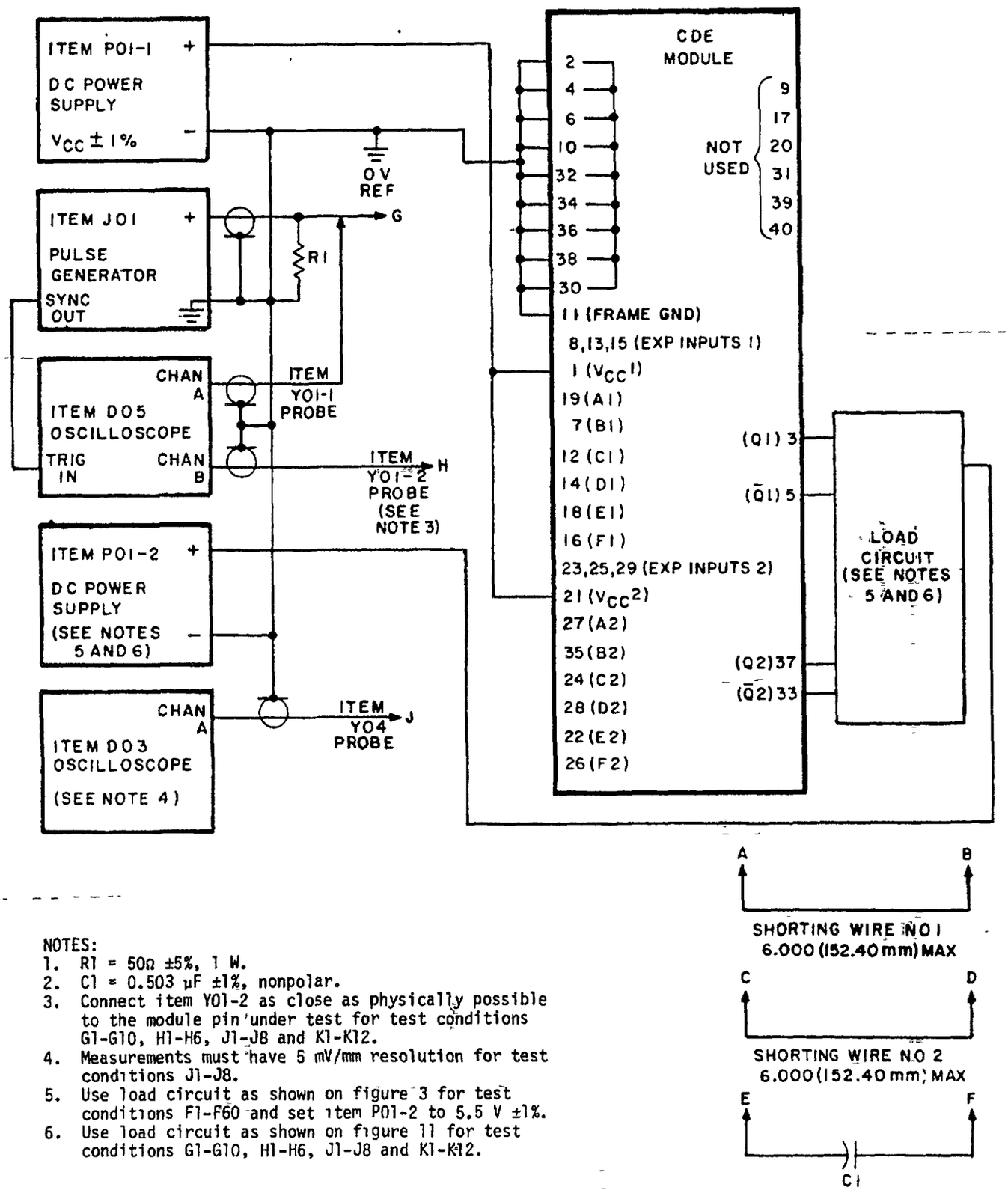
* TABLE X. Output pulse (overshoot, undershoot, preshoot) test.

| Test condition | Parameter | Connect 1/ | | Initial limits 25°±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | G to pin | H to pin | Min | Max | | |
| K1 | Q1 Output pulse, overshoot | 19 | 3 | 0 | 500 | mV | 9, 10 |
| K2 | $\overline{Q1}$ | 19 | 5 | | | | |
| K3 | Q2 | 27 | 37 | | | | |
| K4 | $\overline{Q2}$ | 27 | 33 | | | | |
| K5 | Q1 Output pulse, undershoot | 19 | 3 | | | | |
| K6 | $\overline{Q1}$ | 19 | 5 | | | | |
| K7 | Q2 | 27 | 37 | | | | |
| K8 | $\overline{Q2}$ | 27 | 33 | | | | |
| K9 | Q1 Output pulse, preshoot | 19 | 3 | | | | |
| K10 | $\overline{Q1}$ | 19 | 5 | | | | |
| K11 | Q2 | 27 | 37 | | | | |
| K12 | $\overline{Q2}$ | 27 | 33 | | | | |

1/ Connect pin 12 to pin 16 and pin 24 to pin 26 for all test conditions.



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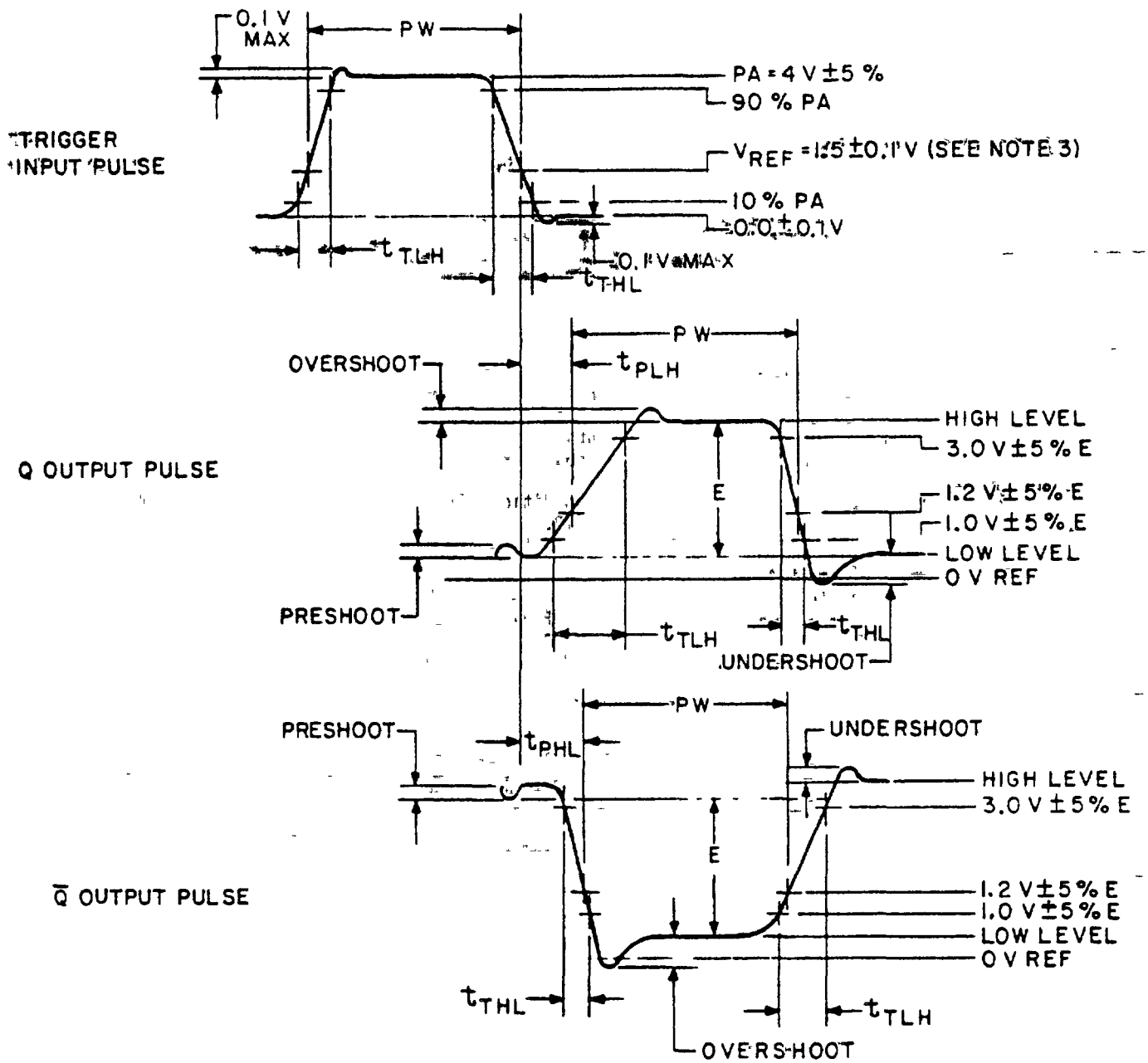



NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. C1 = 0.503 µF ±1%, nonpolar.
3. Connect item Y01-2 as close as physically possible to the module pin under test for test conditions G1-G10, H1-H6, J1-J8 and K1-K12.
4. Measurements must have 5 mV/mm resolution for test conditions J1-J8.
5. Use load circuit as shown on figure 3 for test conditions F1-F60 and set item P01-2 to 5.5 V ±1%.
6. Use load circuit as shown on figure 11 for test conditions G1-G10, H1-H6, J1-J8 and K1-K12.

* FIGURE 9. Output pulse width, transition time ($t_{THL}$ and $t_{TLH}$), propagation delay time ($t_{PLH}$ and $t_{PHL}$), output pulse amplitude (low-level and high-level), and output pulse (overshoot, undershoot, preshoot) test circuit for tables VI through X.



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* FIGURE 10. Output pulse width, transition time ($t_{THL}$ and $t_{TLH}$), propagation delay time ($t_{PLH}$ and $t_{PHL}$), output pulse amplitude (low-level and high-level), output pulse (overshoot, undershoot, preshoot), and input trigger sensitivity and immunity test circuit waveforms for tables VI through X and XII.



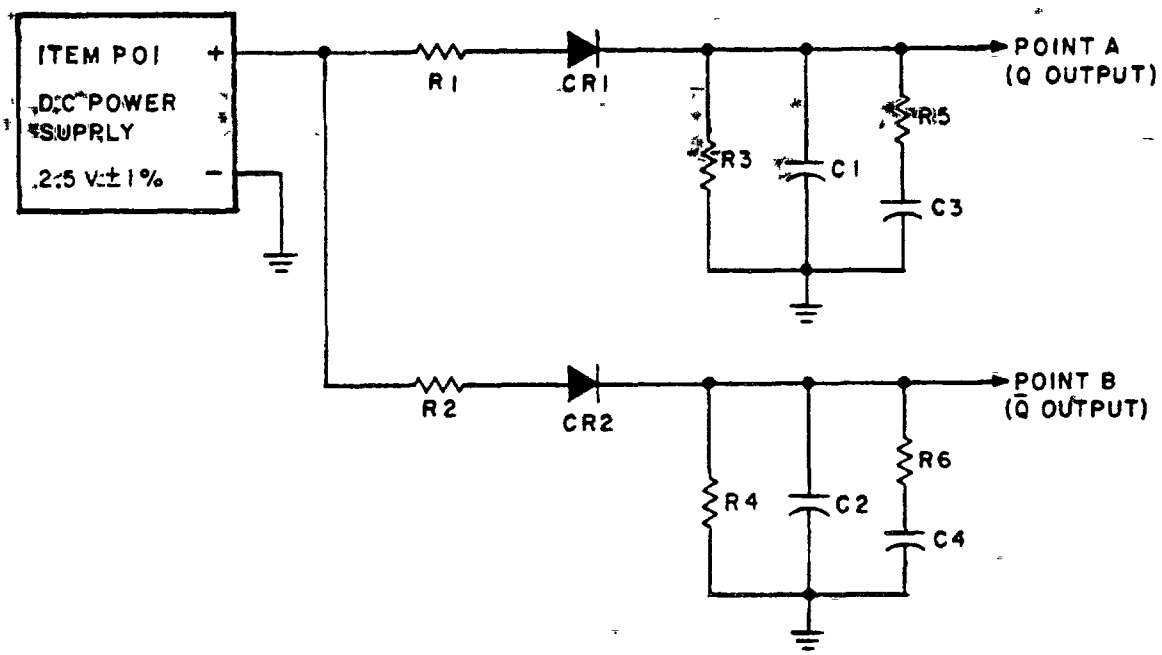
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NOTES:
1. Input pulse width is measured at 50% E for test conditions M1 through M12.
2. Several perturbations which may occur on the output waveform are shown (preshoot, overshoot, undershoot). Perturbations may occur at other locations on the output waveforms, but they must conform to the following requirements: No negative perturbations on the high level or positive perturbations on the low level shall exceed 0.5 V amplitude.
3. Input pulse parameters:
   PW (1.5 V level) = 125 ns ±10% for test conditions F1-F8, F13-F16, F29-F32, F57-F60, G1-G10, H1-H6, J1-J8, and K1-K12.
   = 100 µs ±10% for test conditions F9-F12, F17-F28, and F33-F56.
   $t_{TLH} = t_{THL}$ = 25 ±5 ns.
   PRP = 30 ms ±10% for test conditions F1-F4, G5, G10, H5, and H6.
   = 3 ms ±10% for test conditions F5-F8, F13-F16, F29-F32, F57-F60, G1-G4, G6-G9, H1-H4, J1-J8, and K1-K12.
   = 2 ms ±10% for test conditions F9-F12, F17-F28, and F33-F56.

* FIGURE 10. Output pulse width, transition time ($t_{THL}$ and $t_{TLH}$), propagation delay time ($t_{PLH}$ and $t_{PHL}$), output pulse amplitude (low-level and high-level), output pulse level set, undershoot, preshoot, and input trigger sensitivity and immunity test circuit waveforms for tables VI through X and XII - Continued.



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NOTES:
1. R1 = Select value for a current flow of 35.2 mA ±1% into point A when point A is held at 0.4 V.
2. R2 = Select value for a current flow of 22.4 mA ±1% into point B when point B is held at 0.4 V.
3. This load shall be applied to each module circuit Q and $\bar{Q}$ output.
4. R3 = R4 = 18.7 kΩ ±1%, 1/8 W.
5. R5 = R6 = 100Ω ±1%, 1/8 W.
6. C1 = C2 = 33 pF ±10%.
7. C3 = C4 = 330 pF ±10%.
8. CR1 = CR2 = 1N4150 or equivalent.
9. The lead length from the load circuit to the module pin under test shall be less than 2 inches (50.8mm).

* FIGURE 11. CDE module load circuit.

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* TABLE XI. Output pulse width (recovery time) test.

| Test condition | Parameter | Connect | | | | Initial limits 25°C ±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| L1 | Q1 Output pulse width (recovery time) | 12 | 16 | 19 | 3 | 93 | 107 | µs | 12 |
| L2 | Q1 | Open | Open | 19 | 3 | 1,395 | 1,605 | | |
| L3 | Q2 | 24 | 26 | 27 | 37 | 93 | 107 | | |
| L4 | Q2 | Open | Open | 27 | 37 | 1,395 | 1,605 | | |



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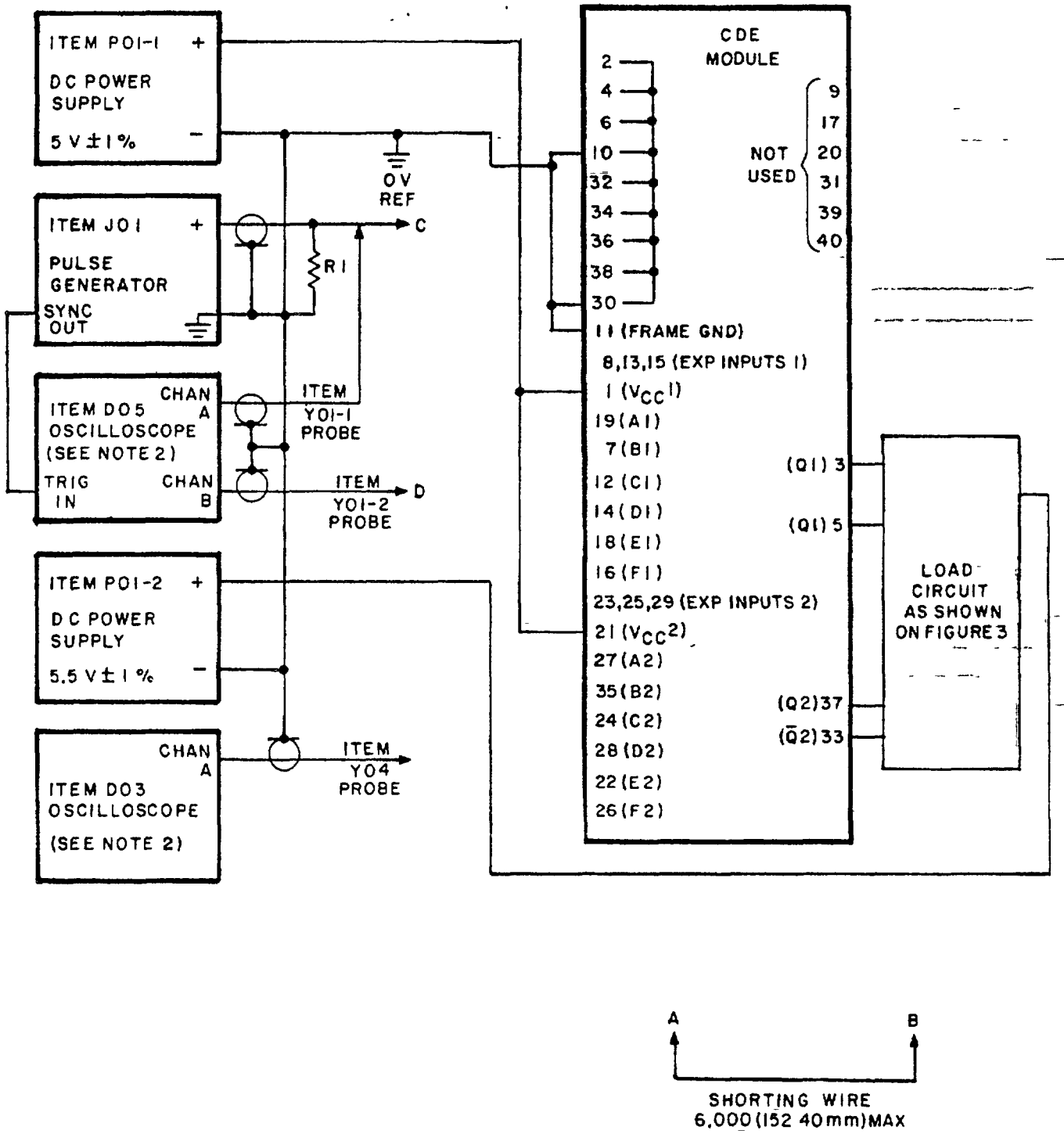
* TABLE XII. Input trigger sensitivity and immunity test:

| Test condition | Parameter | Connect | | | | Initial limits 25°±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | C to pin | D to pin | Min | Max | | |
| M1 | Q1 Input trigger sensitivity | 12 | 16 | 19 | 3 | 93 | 107 | µs | 10, 12 |
| M2 | $\overline{Q1}$ | 12 | 16 | 19 | 5 | | | | |
| M3 | Q2 | 24 | 26 | 27 | 37 | | | | |
| M4 | $\overline{Q2}$ | 24 | 26 | 27 | 33 | | | | |
| M5 | Q1 Input immunity (20 ns) | 12 | 16 | 19 | 3 | 0 | 5 | ns | |
| M6 | $\overline{Q1}$ | 12 | 16 | 19 | 5 | | | | |
| M7 | Q2 | 24 | 26 | 27 | 37 | | | | |
| M8 | $\overline{Q2}$ | 24 | 26 | 27 | 33 | | | | |
| M9 | Q1 Input immunity (0.95 V) | 12 | 16 | 19 | 3 | | | | |
| M10 | $\overline{Q1}$ | 12 | 16 | 19 | 5 | | | | |
| M11 | Q2 | 24 | 26 | 27 | 37 | | | | |
| M12 | $\overline{Q2}$ | 24 | 26 | 27 | 33 | | | | |





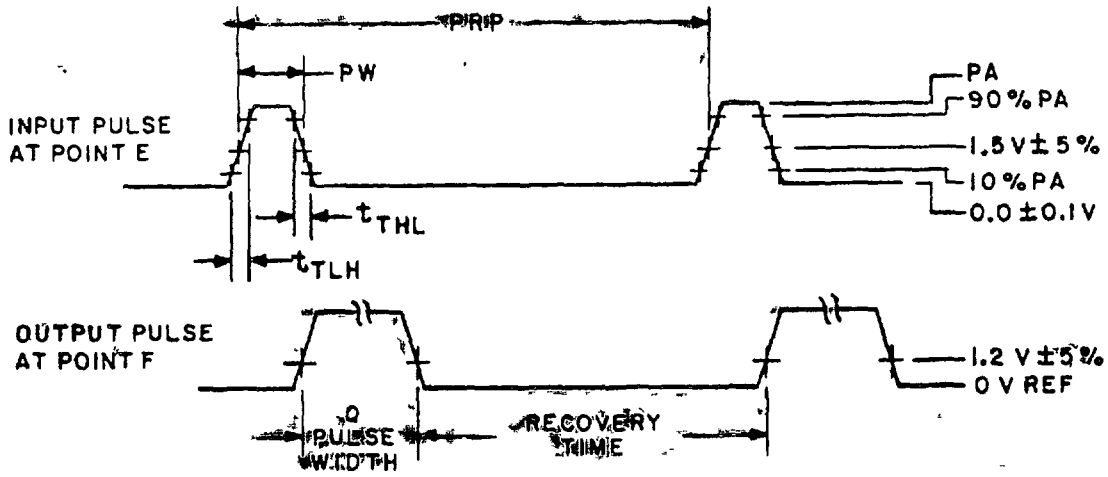
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* FIGURE 12. Output pulse width (recovery time) and input trigger sensitivity and immunity test circuit for tables XI and XII.



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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. The item J01 pulse amplitude, pulse width, and transition times are adjusted using item D05 and item Y01. The item J01 repetition period and the module Q pulse width are measured using items D03 and Y04.
3. Input pulse parameters (test condition L1).
   PA (reference to 0 V) = 4 V ±5%.
   PW (1.5 V level) = 125 ns ±10%.
   $t_{TLH} = t_{THL}$ = 20 ns ±10%.
   PRP = 360 µs ±5%.
4. Input pulse parameters (test condition L2):
   PA (reference to 0 V) = 4 V ±5%.
   PW (1.5 V level) = 125 ns ±10%.
   $t_{TLH} = t_{THL}$ = 20 ns ±10%.
   PRP = 1760 µs ±5%.
5. Input pulse parameters (test condition L3):
   PA (reference to 0 V) = 4 V ±5%.
   PW (1.5 V level) = 125 ns ±10%.
   $t_{TLH} = t_{THL}$ = 20 ns ±10%.
   PRP = 3.8 ms ±5%.
6. Input pulse parameters (test conditions L4):
   PA (reference to 0 V) = 4 V ±5%.
   PW (1.5 V level) = 125 ns ±10%.
   $t_{TLH} = t_{THL}$ = 20 ns ±10%.
   PRP = 17.8 ms ±5%.
7. Input pulse parameters (test conditions M1 through M4):
   PA (reference to 0 V) = 3.2 V ±5%.
   PW (1.5 V level) = 135 ns ±5%.
   $t_{TLH} = t_{THL}$ = 40 ns ±5%.
   PRR = 100 kHz ±10%.
8. Input pulse parameters (test conditions M5 through M8):
   PA (reference to 0 V) = 5.0 V ±5%.
   PW (1.5 V level) = 20 ns ±5%.
   $t_{TLH} = t_{THL}$ = 8 ns ±5%.
   PRR = 100 kHz ±10%.
9. Input pulse parameters (test conditions M9 through M12):
   PA (reference to 0 V) = 0.95 V ±5%.
   PW (1.5 V level) = 1000 ns ±5%.
   $t_{TLH} = t_{THL}$ = 8 ns ±5%.
   PRR = 100 kHz ±10%.

* FIGURE 12. Output pulse width (recovery time) and input trigger sensitivity and immunity test circuit for tables XI and XII - Continued.



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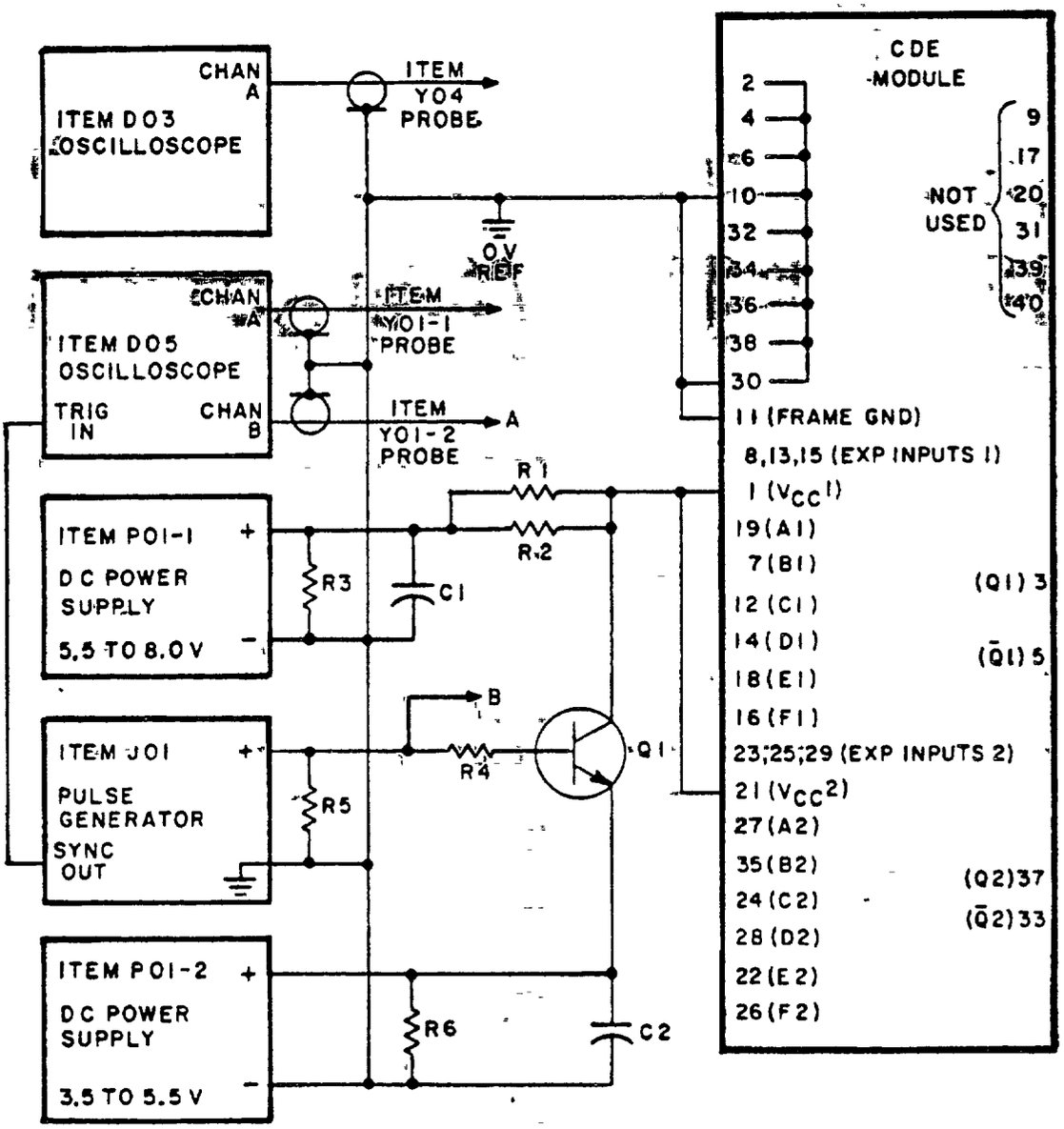
* TABLE XIII. Power supply transient (rise time and false trigger) test.

| Test condition | Parameter | Connect A to pin | Initial limits 25°±5°C ambient Min | Initial limits 25°±5°C ambient Max | Unit | Test figure |
|---|---|---|---|---|---|---|
| N1 | $\overline{Q1}$ Power supply transient rise time | 5 | 9 | 53 | µs | 13 |
| N2 | $\overline{Q2}$ | 33 | 9 | 53 | ↓ | ↓ |
| N3 | $\overline{Q1}$ | 5 | 1.5 | 40.0 | ↓ | |
| N4 | $\overline{Q2}$ | 33 | 1.5 | 40.0 | ↓ | |
| N5 | $\overline{Q1}$ Power supply transient false trigger | 5 | 2.0 | 5.5 | V | |
| N6 | $\overline{Q2}$ | 33 | ↓ | ↓ | ↓ | ↓ |
| N7 | $\overline{Q1}$ | 5 | ↓ | ↓ | ↓ | 14 |
| N8 | $\overline{Q2}$ | 33 | ↓ | ↓ | ↓ | 14 |





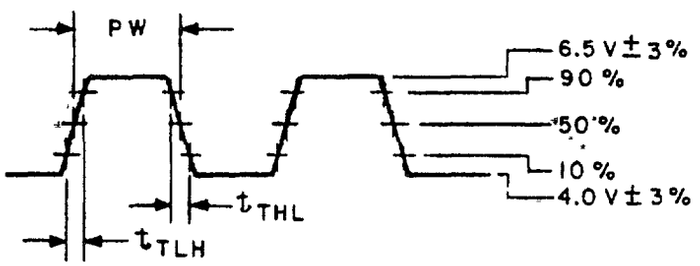
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* FIGURE 13. Power supply transient (rise time and false trigger) test circuit for table XIII.

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NOTES:
1. R1 = R2 = 10Ω ±1%, 1 W.
2. R3 = R6 = 18.2Ω ±1%, 3 W.
3. R4 = R5 = 51.1Ω ±1%, 1/2 W.
4. C1 = C2 = 1.0 µF ±10%.
5. Q1 = 2N3108A or equivalent.



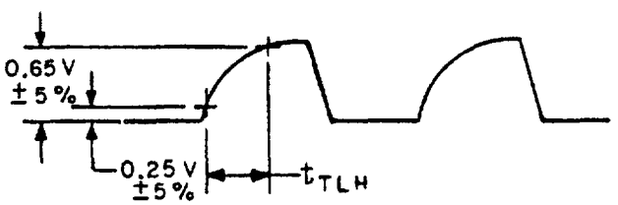

6. Using item D03, adjust item J01 to obtain the preceding waveform at point B with the output of item J01 disconnected from R4.

$t_{TLH} = t_{THL}$ = 50 ns ±20%.
PW (at 50% amplitude) = 400 µs ±10%.
Duty cycle = 50%.




VOLTAGE PULSE AT MODULE PIN 1 (MODULE PLUGGED IN)

7. For test conditions N1 through N6; connect item J01 to R4; connect item Y04 to module pin 1; and adjust item P01-2 to set the baseline of the voltage pulse at module pin 1 (module plugged in) to 4.5 V ±3%. Then adjust item P01-1 to obtain a peak to peak amplitude of 0.9 V ±5%.




OUTPUT PULSE

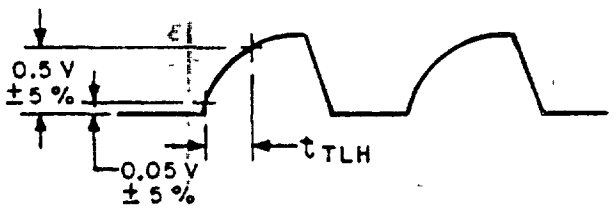
8. For test conditions N1 through N2, the output pulse rise time shall be measured as shown above and shall conform to the requirements of table XIII using item D05 and item Y01-2 (with 1X attenuator and AC coupler).

* FIGURE 13. Power supply transient (rise time and false trigger) test circuit for table XIII - Continued.



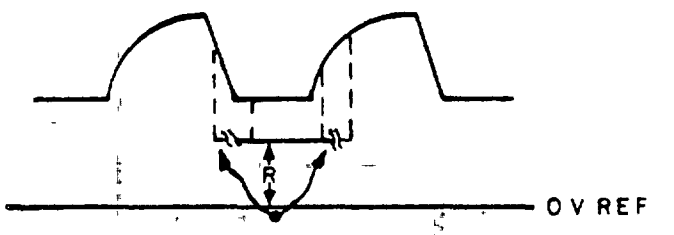
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9. For test conditions N3 and N4, the output pulse rise time shall be measured as shown below and shall conform to the requirements of table XIII using item D05 and item Y01-2 (with 1X attenuator and AC coupler).




OUTPUT PULSE

10. For test conditions N5 and N6, the output pulse measurement (R) shall conform to the requirements of table XIII using item D03. The value R is defined as the minimum voltage of the waveform.



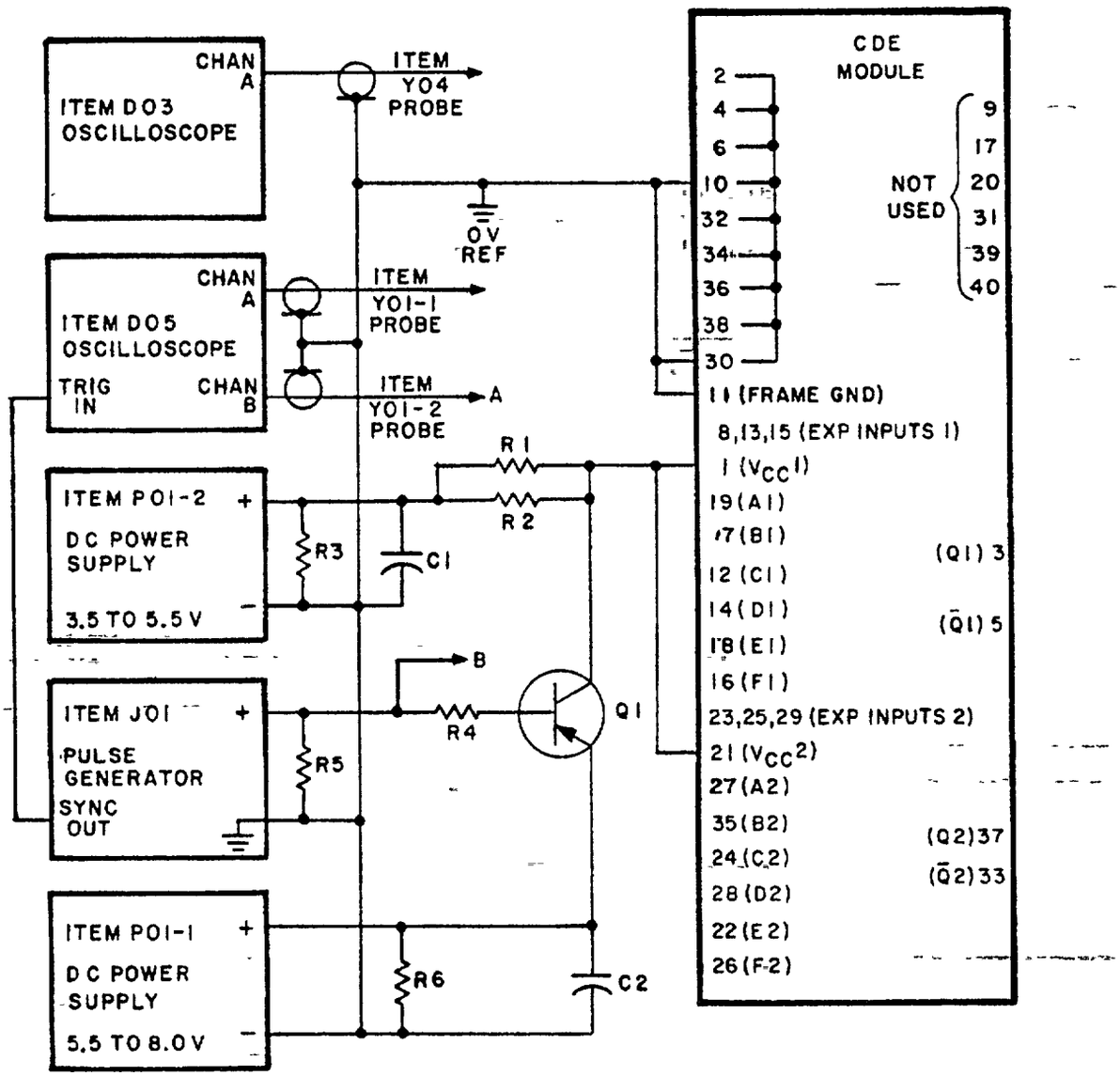

MODULE FALSE TRIGGERING MAY OCCUR AT EITHER OF THESE TWO POINTS

OUTPUT PULSE

* FIGURE 13. Power supply transient (rise time and false trigger) test circuit for table XIII - Continued.



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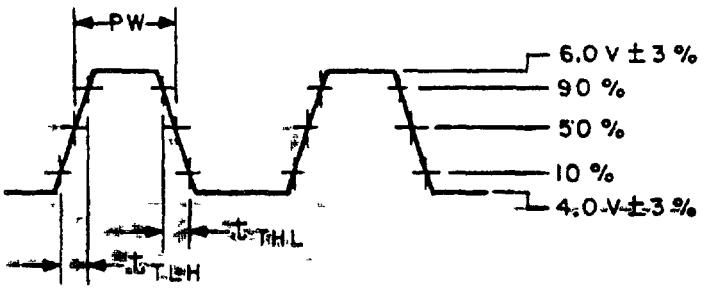



* FIGURE 14. Power supply transient false trigger test circuit for table XIII.

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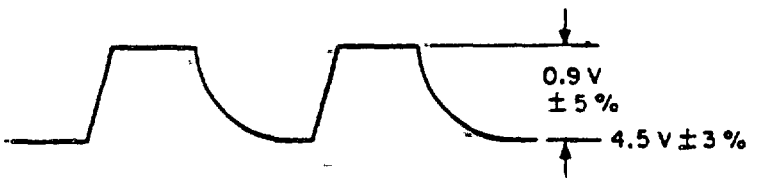
NOTES:

1. R1 = R2 = 10Ω ±1%, 1 W.
2. R3 = R6 = 18.2Ω ±1%, 3 W.
3. R4 = R5 = 51.1Ω ±1%, 1/2 W.
4. C1 = C2 = 1.0 µF ±10%.
5. Q1 = 2N4029 or equivalent.



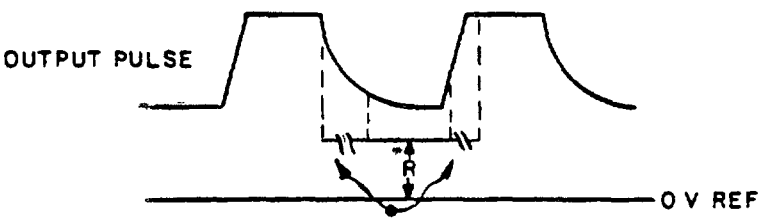

6. Using item D03, adjust item J01 to obtain the preceding waveform at point B with the output of item J01 disconnected from R4.

   $t_{TLH} = t_{THL}$ = 50 ns ±10%.
   PW (at 50% amplitude) = 400 µs ±10%.
   Duty cycle = 50%.




VOLTAGE PULSE AT MODULE PIN 1 (MODULE PLUGGED IN)

7. For test conditions N7 and N8, connect item J01 to R4; connect item Y04 to module pin 1; and adjust item P01-2 to set the baseline of the voltage pulse at module pin 1 (module plugged in) to 4.5 V ±3%. Then adjust item P01-1 to obtain a peak to peak amplitude of 0.9 V ±5%.




MODULE FALSE TRIGGERING MAY OCCUR AT EITHER OF THESE TWO POINTS

8. For test conditions N7 and N8, the output pulse measurement (R) shall conform to the requirements of table XIII using item D03. The value R is defined as the minimum voltage of the waveform.

* FIGURE 14. Power supply transient false trigger test circuit for table XIII - Continued.



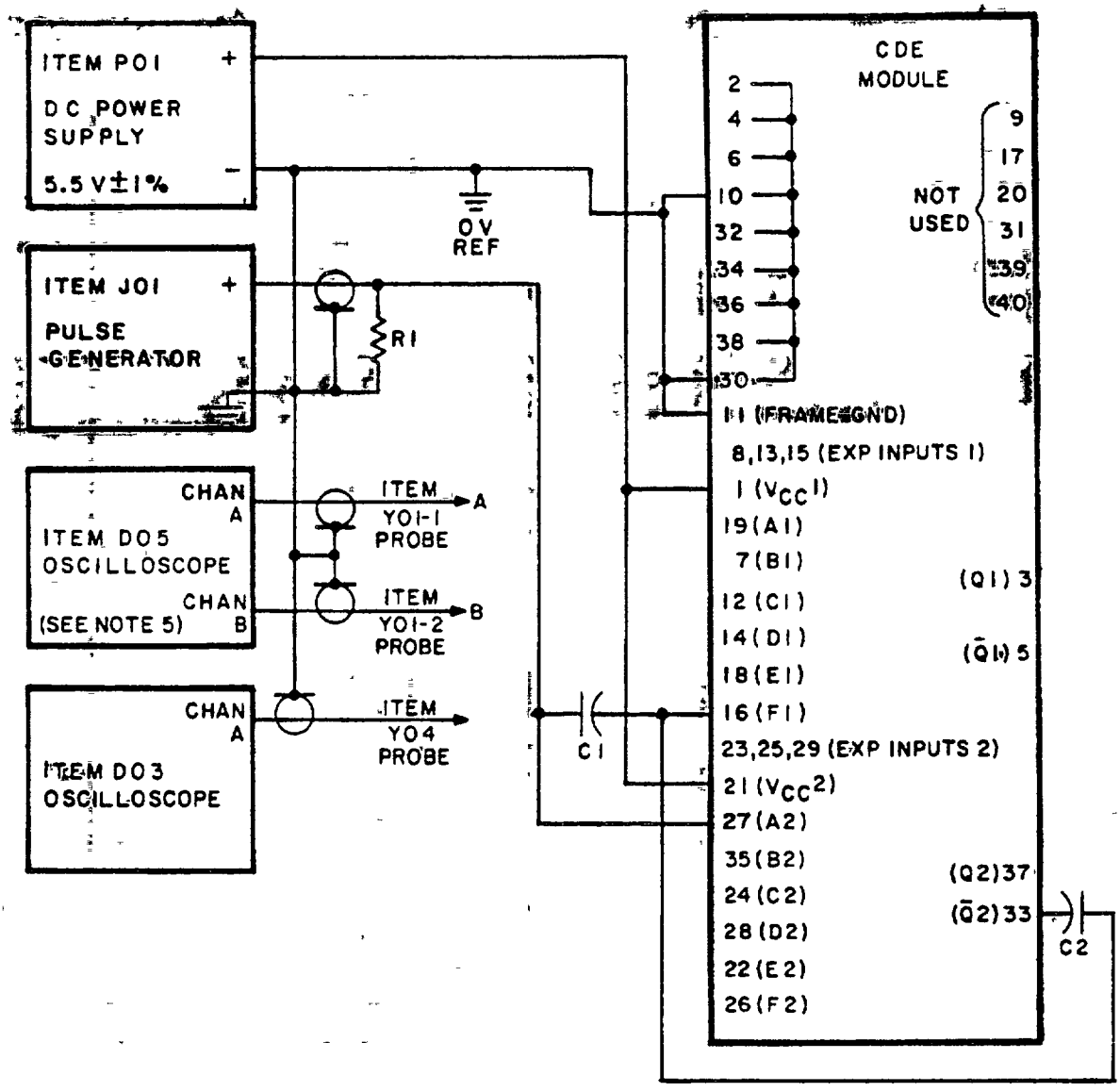
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* TABLE XIV. Crosstalk test.

| Test condition | Parameter | Connect | | Initial limits 25°±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|
| | | A to pin | B to pin | Min | Max | | |
| P1 | Q1 Crosstalk | 27 | 3 | 0 | 5 | ns | 15,17 |
| P2 | Q1 | 33 | 3 | | | | |
| P3 | Q1 | 33 | 3 | | | | |
| P4 | $\overline{Q1}$ | 27 | 5 | | | | |
| P5 | $\overline{Q1}$ | 33 | 5 | | | | |
| P6 | $\overline{Q1}$ | 33 | 5 | | | | |
| P7 | Q2 | 19 | 37 | | | | 16,17 |
| P8 | Q2 | 5 | 37 | | | | |
| P9 | Q2 | 5 | 37 | | | | |
| P10 | $\overline{Q2}$ | 19 | 33 | | | | |
| P11 | $\overline{Q2}$ | 5 | 33 | | | | |
| P12 | $\overline{Q2}$ | 5 | 33 | | | | |



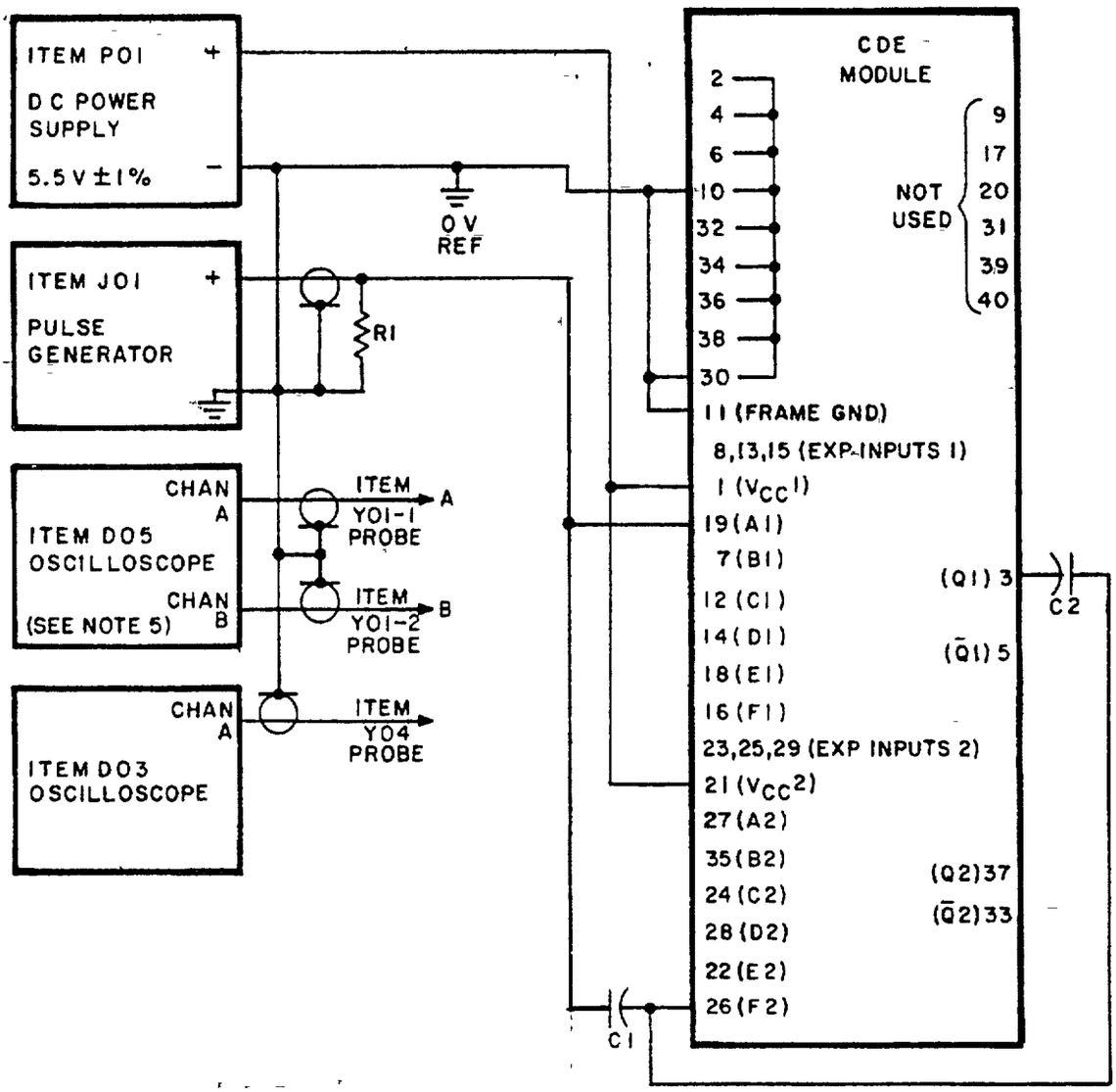
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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. C1 = 4 pF maximum ±10%.
3. C2 = 9 ±1 pF.
4. The capacitances are measured using item C02 at the test fixture pins with the module removed, at a frequency of 1 MHz ±10%, and will include stray wiring capacitance but not oscilloscope probe capacitance. Glass trim capacitors may be added to conform to the required value.
5. Set item D05 scope internal trigger slope to negative for test conditions P1, P3, P4, and P6 and to positive for test conditions P2 and P5.

* FIGURE 15. Crosstalk test circuit for table XIV.

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NOTES:
1. R1 = 50Ω ±5%, 1 W.
2. C1 = 4 pF maximum ±10%.
3. C2 = 9 ±1 pF.
4. The capacitances are measured using item C02 at the test fixture pins with the module removed, at a frequency of 1 MHz ±10%, and will include stray wiring capacitance but not oscilloscope probe capacitance. Glass trim capacitors may be added to conform to the required values.
5. Set item D05 scope internal trigger slope to negative for test conditions P7, P9, P10 and P12 and to positive for test conditions P8 and P11.

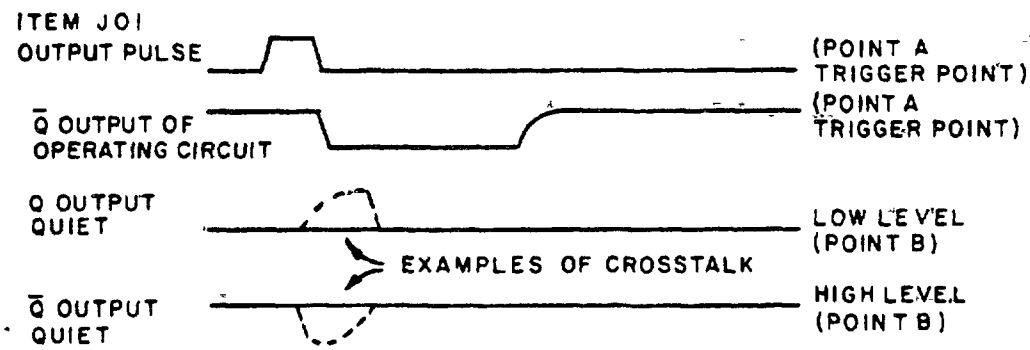
* - FIGURE 16. Crosstalk test circuit for table XIV.

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NOTES:
1. Using item D05, adjust the output of item J01 to obtain the following input pulse:

   $V_{PA}$ = 4 V ±5%.
   PW (2 V level) = 125 ns ±10%.
   $t_{TLH} = t_{THL}$ = 10 ns ±10%.
   PRP = 2 ms ±10% (use item D03 with item Y04 for this measurement).

2. To examine the output waveforms under test, item D05 shall be internally triggered using the signal on channel A (probe Y01-1).




3. For each test condition, the maximum pulse width detected at point B, measured at 1.0 V above the low level on the Q output and at 3.0 V below the high level on the $\overline{Q}$ output, shall not exceed the limits of table XIV.

* FIGURE 17. Crosstalk test circuit waveforms for table XIV.



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* TABLE XV. Zener applied potential and trigger voltage high test.

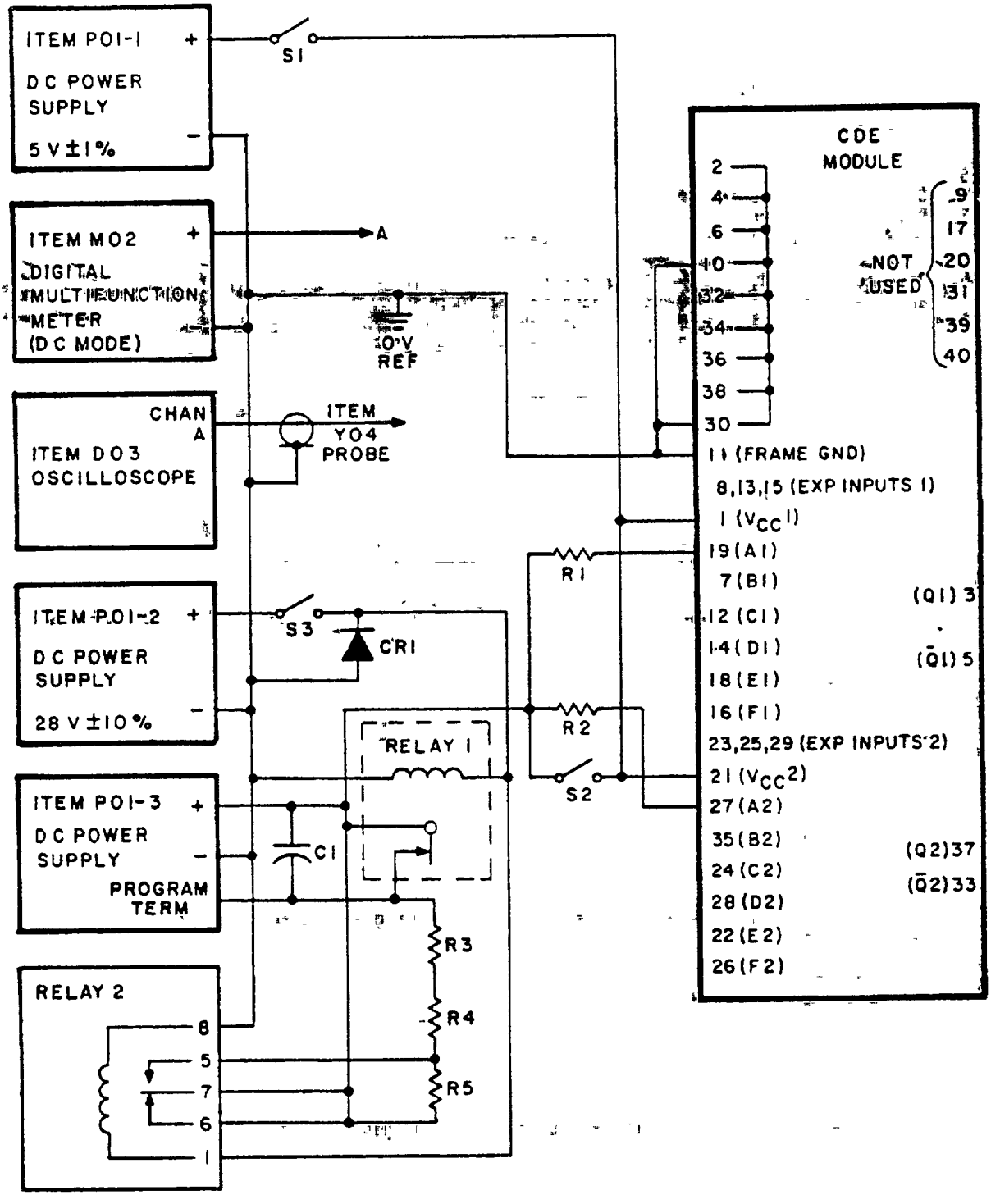
| Test condition | Parameter | S1 | S2 | Connect | Initial limits 25°±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|---|---|
| | | | | A to pin | Min | Max | | |
| Q1 | $\overline{Q1}$ Zener applied potential | Open | Closed | 5 | 3.8 | 5.5 | V | 18 |
| Q2 | $\overline{Q2}$ Zener applied potential | Open | Closed | 33 | 3.8 | 5.5 | V | |
| Q3 | Q1 Trigger voltage high | Closed | Open | 3 | 0 | 400 | mV | |
| Q4 | Q2 Trigger voltage high | Closed | Open | 37 | 0 | 400 | mV | |

* TABLE XVI. $V_{OUT}$ (low-loaded) test.

| Test condition | Parameter | Connect | Initial limits 25°±5°C ambient | | Unit | Test figure |
|---|---|---|---|---|---|---|
| | | A to pin | Min | Max | | |
| R1 | Q1 $V_{OUT}$ (low loaded) | 3 | 0.0 | 1.1 | V | 19 |
| R2 | Q2 $V_{OUT}$ (low loaded) | 37 | 0.0 | 1.1 | V | 19 |



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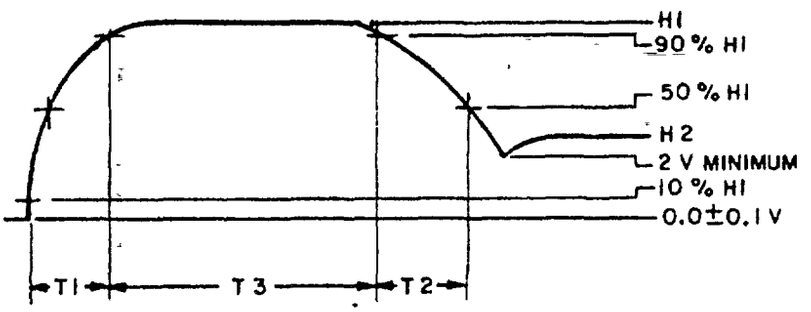



* FIGURE 18. Zener applied potential and trigger voltage high test circuit for table XV.



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NOTES:
1. R1 = R2 = 154Ω ±1%, 1/8 W.
2. R3 = 178Ω ±1%, 1/8 W.
3. R4 = 301Ω ±1%, 1/8 W.
4. R5 = 649Ω ±1%, 1/8 W.
5. C1 = 22 µF ±10%.
6. CR1 = 1N647 or equivalent.
7. Relay 1 = 28 V relay.
8. Relay 2 = Agastat model 2112 or equivalent.
9. S1 = S2 = S3 = SPST switch.
10. Resistors R3, R4, R5, and capacitor C1 are programming components for P01-3 and are based on a programming constant of 100Ω per volt.
11. Using item D03 with item Y04, adjust the time delay relay to obtain the following output waveform at pin 7 of the time delay relay after switch S3 is closed.




Waveform parameters:

H1 = 11.25 V ±1%.
H2 = 5.0 V ±10%.

T1 = 100 $^{+100}_{-\ 50}$ ms.

T2 = 100 $^{+400}_{-\ 50}$ ms.

T3 = 0.5 s ±10%.

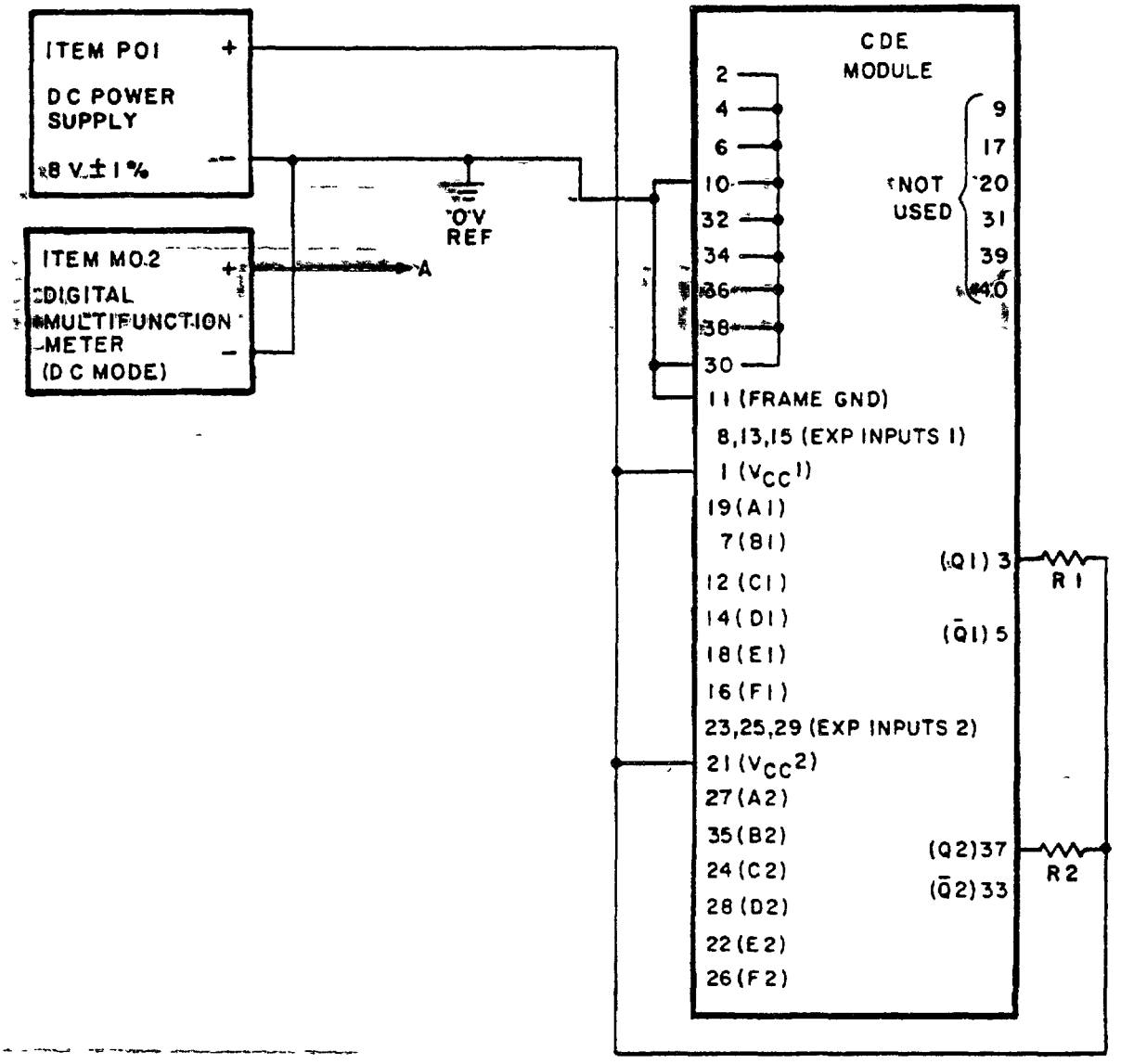
12. Item P01-3 should be selected to ensure that the programming terminal is floating (-).
13. Test sequence:

a. Close switch S3.
b. Wait 2 seconds minimum.
c. Make voltage measurements for test conditions.
d. Open switch S3.

* FIGURE 18. Zener applied potential and trigger voltage high test circuit for table XV - Continued.



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NOTES:
1. R1 = R2 = 82Ω ±5%, 3 W.
2. The measurement for each output shall be made after a time period of at least 10 seconds.

* FIGURE 19. $V_{OUT}$(low-loaded) test circuit for table XVI.



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